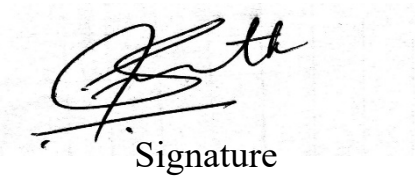
A Natural Language Processing with Text Mining Approach for Fraud Detection in Social Media Platforms.

Thesis for the partial fulfilment of the requirements for the Masters Degree in MSc Information Technology.

DECLARATION OF ORIGINALITY

I declare that this is an original study based on my own work and that I have not submitted it for any other course or degree.



Signature

Date: 15 Dec 2023

ABSTRACT

This master's research project aims to address the significant challenges in identifying fraudulent activities inside the ever-changing environment of social media. The primary goals are to explore the usefulness of NLP (Natural Language Processing) and approaches, such as text mining, Naive Bayes, TF-IDF and N-gram modelling in detecting fraudulent patterns. The research explores the contextual intricacies and ethical factors that are essential to the process of detecting fraud, considering the changing dynamics of social media interactions. The research examines three discrete datasets—cyberbullying, fraudulent reviews, and misinformation or rumours regarding disaster as the metrics for assessing the adaptability of fraud detection approaches.

In addition to Naive Bayes, N-gram modelling, and TF-IDF, the study used Unigram and Bigram techniques with addition to their combined method, to further optimize the process of feature extraction. The purpose of the study was to improve the system's capability to identify fraudulent actions by using a more refined knowledge of language use. The algorithm was able to recognize fraudulent patterns across a variety of levels within the various Twitter datasets that were being investigated employing different feature extraction method that helped to the development of a powerful fraud detection framework.

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LIST OF ACRONYMS

AFF	–	Adaptive Feature Fusion
ANN	–	Artificial Neural Networks
BoW	–	Bag-of-Words
DT	–	Decision Tree
IE	–	Information Extraction
IR	–	Information Retrieval
KNN	–	K-Nearest Neighbor
LR	–	Linear Regression
ML	–	Machine Learning
NLP	–	Natural Language Processing
OCSVM	–	One-Class Support Vector Machine
PCA	–	Principal Component Analysis
QA	–	Question-Answering
RF	–	Random Forest
SNA	–	Social Network Analysis
SVM	–	Support Vector Machine
TF-IDF	–	Term Frequency-Inverse Document Frequency
URL	–	Uniform Resource Locator
WEKA	–	Waikato Environment for Knowledge Analysis
WWW	–	World Wide Web

CHAPTER ONE: INTRODUCTION

1.1. Background of the Research

Our daily lives depend on social media for communication, knowledge exchange, and community centres. Social media are a important online platform for social networking where users may exchange their thoughts and opinions on media content (Sharmin and Zaman, 2017). However, as these platforms have grown in popularity, there has been an increase in fraudulent activities such as frauds, false accounts, disinformation, and other deceptive practises. Fraudulent individuals take use of social media's openness and interconnectedness to achieve a variety of harmful goals, such as financial gain, reputation damage, or the distribution of fake information. Social media fraud includes phishing, identity theft, scams, fake reviews, click fraud, data breaches, spam, cyberbullying, deepfake content, and account takeovers, compromising user trust and privacy.

Fraud detection in social media is a critical action aimed at protecting users from misleading actions and preserving the integrity of these platforms. According to Price Waterhouse Coopers research, 30% of the organisations examined have already been victims of fraud (Sánchez-Aguayo et al., 2021). The large amount of data collected, the complexity of fraudulent activity, and the shifting techniques adopted by hostile players make identifying fraud in this scenario difficult.

NLP along with ML is the current a mutually beneficial combination for detecting fraud and misread data (Putatunda, 2020). The application of this field includes the manipulation and examination of a great deal of natural language data through combining the fields of languages, computer science, and artificial intelligence. The ultimate goal is to create software that can comprehend text and speech within their respective contexts and meanings. A machine learning model learns the rules used to solve problems by analysing different aspects of language, such as syntax, semantics, pragmatics, and morphology. NLP is an area of AI (Artificial Intelligence) that studies the interplay of computers and human language when applied to social media, allows for the analysis of textual material in posts, comments, and profiles, revealing significant information about user behaviour and intents.

Text mining, on the other hand, entails extracting meaningful information from unstructured text data. When analysing massive amounts of data for fraud research, text mining has become more

and more popular using big data resources This makes it possible for a lot of businesses to find critical information in customer reviews, corporate documents, social media posts, emails, phone logs, and more. It also helps them spot fraud, dangers, and client attrition (Pejić Bach et al., 2019). Text cleaning, tokenization, sentiment analysis, and feature extraction are among the procedures covered. Combining NLP and text mining with other data analysis approaches like as machine learning, anomaly detection, and network analysis results in a holistic strategy to detecting and mitigating fraud on social media platforms.

This combined strategy allows professionals in the field to detect existing fraudulent behaviours while also anticipating upcoming threats and staying one step ahead of hostile actors. It enables the development of algorithms and systems capable of detecting suspicious behaviour, patterns, and connections on their own, thus protecting users and maintaining trust in the social media ecosystem.

1.2. Problem Statement

Social media is an excellent venue for scammers since huge number people are engaged in social media platforms. According to survey in 2017, they are used by three billion people, or 40% of the world population, with a million new members added every day (Kemp, 2017). Thus, extensive use of social media platforms nowadays has given birth to a large and rising threat. Fraudulent actions such as creation of false accounts, the propagation of disinformation, different types of frauds operations threatening users, the reliability of information provided on these platforms, and the integrity of the platforms itself are most of them. Detecting and combating social media fraud is a complicated and changing task that necessitates modern methodologies and tools. The following major issues are addressed in this study:

1. Identification of Emerging Fraud Patterns: Social media platforms are constantly evolving environment in which fraudsters constantly adapt and create new strategies. It is critical to recognise and comprehend developing fraud patterns and strategies employed by fraudulent users on these platforms.

2. Efficient and Accurate Fraud Detection: It is critical to develop efficient and accurate fraud detection techniques. The present methods usually depend on human reporting or simple rule-based systems, resulting in long response times and high false-positive rates. The goal is to develop technology that can detect fraudulent activity quickly and correctly.

3. User Privacy and Ethical Considerations: When implementing fraud detection techniques, users' privacy and ethical norms must be respected. Finding an acceptable balance between

effective fraud detection and preserving consumers' rights and privacy is a difficult and continuing task.

4.Interdisciplinary Approach: An interdisciplinary strategy combining NLP, text mining, (ML) machine learning, and behavioural analysis is required to handle social media fraud thoroughly. This strategy should evolve with the fraud environment and cover a wide spectrum of fraudulent actions.

5. Legal and Regulatory Compliance: Detecting fraud on social media platforms and within the boundaries of relevant legal and regulatory frameworks is a critical challenge. Navigating data protection laws, user rights, and privacy regulations is a complex and time-consuming task.

1.3. Research Question

How can NLP techniques with text mining be used to extract meaningful features from social media activities for fraud detection and what are its effectiveness for the detection?

1.4. Research Aim

The main goal of the project is to create a fraud detection model that combines text mining and NLP methods to identify fraudulent activity on social media platforms with a high degree of efficiency and accuracy. This research typically works on two kinds of fraud that are detecting fake information and cyberbullying.

However, this research aims to investigate these problems by developing innovative method, algorithm, and system that leverage NLP and text mining technique:

1. The study would develop an advanced and adaptive fraud detection system for social media platforms that leverages NLP and text mining techniques.
2. The study would also enhance the understanding of emerging fraud patterns and tactics employed by malicious actors on social media specially in detecting fake information and cyberbullying.
3. It also respects and uphold user privacy and ethical considerations while implementing those fraud detection mechanisms.
4. This study would foster interdisciplinary collaboration by integrating NLP, text mining, ML, and behavioural analysis in the context of social media fraud detection.
5. It ensures compliance with relevant legal and regulatory frameworks during the research and implementation of fraud detection method.

1.5. Research Objectives

1. Collect, process and analyses historical data to identify and characterize emerging fraud patterns on social media platforms.
2. Develop algorithms that can adapt to evolving fraud tactics and recognize anomalies in user behaviour or content.
3. Create automated fraud detection mechanisms that reduce response times and false-positive rates.
4. Implement machine learning models, which in this case supervised learning techniques has been used to classify and identify fraudulent activities.
5. Develop and apply guidelines for data collection and analysis that respect user privacy and adhere to ethical standards.
6. Combine NLP, text mining, machine learning, and network analysis to provide a comprehensive solution for detecting social media fraud typically in this research fake information and cyberbullying detection.
7. Facilitate collaboration between experts in different domains to develop a holistic approach to those fraud detection.
8. Implement thorough testing and validation to assess the established fraud detection system's performance and efficacy.
9. Evaluate and assess the system's performance using different metrics like accuracy, recall, speed, and precision. Examine the system's performance under various scenarios.
10. Facilitate knowledge transfer by providing practical guidance and tools for social media platforms to enhance their fraud detection capabilities.

1.6. Research Justification

The research justification or rationale should describe the requirements for performing the investigation study based on a detailed evaluation of the domains for analyses. The goal of the research on "Natural Language Processing with Text Mining Approach for Fraud Detection in Social Media Platforms" is to provide an innovative and effective strategy to fraud detection in social media platforms by combining NLP with text mining techniques. Due to their vast and relatively easy-to-access user base, social media sites have become a key target for fraudsters. Fraudsters can use online media platforms to disseminate fake and false information, target susceptible individuals, and conduct fraudulent transactions.

Existing fraud detection methods are frequently ineffective in detecting new and developing fraud tendencies. This is because these solutions are frequently rule-based and require manual changes

to stay up with the ever-changing techniques of fraudsters NLP and text mining are strong techniques that may be used to automate the study of social media data and find patterns and deviations that may suggest fraudulent behaviour. This study attempts to address the following difficulties by providing a unique NLP and text mining technique to fraud detection in social media platforms:

Detecting new and emerging fraud patterns: By monitoring social media data for suspicious activity, NLP and text mining may be utilised to uncover new and emerging fraud trends. This can assist in overcoming the limits of rule-based fraud detection technologies.

Improving fraud detection accuracy and efficiency: NLP and text mining can automate the study of social media data and identify fraudulent behaviour with greater accuracy and efficiency than human techniques.

Reducing the cost of fraud detection: By automating the examination of social media data and reducing the need for manual review, NLP and text mining can assist to minimise the cost of fraud detection.

1.7. Research Significance

The research on "Natural Language Processing with Text Mining Approach for Fraud Detection in Social Media Platforms" having the capacity to significantly effect on society by assisting law enforcement agencies in their investigations, protecting social media users from fraud, reducing business losses, and assisting law enforcement agencies in their investigations.

The research may, for example, lead to the creation of new algorithms for detecting bogus activities like, fake information and cyberbullying which further can be implemented in other activities like, phishing schemes, and fraudulent transactions. This will aid social media sites in better protecting their consumers from such fraud. The study might potentially lead to the creation of new technologies for monitoring social media data for fraudulent behaviour and detecting fraudulent transactions. This would assist firms in reducing losses due to social media fraud.

Finally, this study adds to the Bok (Body of Knowledge) on NLP and text mining for fraud detection. As a result, new and creative strategies for feature extraction, model training, and model assessment in the context of fraud detection would be developed. Hence the study on "Natural Language Processing with Text Mining Approach for Fraud Detection in Social Media Platforms" has the potential to benefit society by assisting in the protection of social media users, companies, and law enforcement agencies against fraud.

CHAPTER TWO: LITERATURE REVIEW

2.1. Introduction: Fraud on social media

In the context of online social networks, frauds are defined as actions that lead to financial loss, harassment, damage to one's reputation, loss of faith in the system or in a particular person (Apte, Palshikar and Baskaran, 2019). Online social network frauds are a subset of cybercrimes and white-collar crimes. The authors Zakia Zaman and Sadia Sharmin (2017) addresses the growing issue of spam and malicious content in online social networks while social media platforms offer users the opportunity to share thoughts and information. They are increasingly vulnerable to misuse by malicious users who post phishing links, advertisements, and fraudulent content in comment sections, potentially leading to the transmission of viruses and malware.

2.1.1. Types of Fraud in social media

Misinformation or Rumours: This is the fraud which involves spreading false information choosing a community or a specific group of people who are most likely to disseminate the content (Wang, Musaev and Pu, 2016). They distribute the data throughout the designated network which results in the cause of unrest and fear in social media and society at large.

Phishing: As per writer Jagatic *et al.* (2007), phishing is the theft and improper use of private and sensitive data by making a copy of a well-known website (such as a bank's login page) and making sure the name of phishing website is very similar to the name of the original. If users are directed to these websites from the links found in posts on social media and through forwarding which can result in the release of private information, loss of money, theft of security credentials or personal information.

Fake Reviews: It is the process of intentionally increase or decrease a product's rating to influence a prospective user's choice. It is done by writing several identical evaluations or give a false impression of the quality of an item. According to Lappas (2012), fake reviews can result a potential user's decision to be influenced by the item's enhanced or damaged reputation because of several intentionally placed, erroneous evaluations.

Cyber Bullying: The goal under cyber bullying is to persistently force and inappropriate conversation or contact on a person who is susceptible. This is typically done by obtaining access to the target's trusted network and steal someone therein by adopting a phony identity and monitor the target in secret. They use false identities to send offensive or threatening messages to the target

to intimidate them (Dreßing et al., 2013). They at the end blackmail, intimidation, or harassment of the target.

2.2. Natural Language Processing (NLP)

The term NLP which stands as "natural language processing" refers to a collection of computing approaches that are based on theory and are used to analyse and represent texts that occur naturally at one or more levels of language analysis. The goal of NLP is to achieve language processing like a human for a variety of tasks or applications. In the concept introduced by Chowdhary (2020), NLP is a collection of motivated computer methods which is theoretically automatic evaluation and presentation of human languages. NLP enables users to ask queries on any subject and get prompt answers. With the use of this technique, computers can do a variety of language-related jobs and interact with people in their native tongue. The three main challenges associated with NLP are typically language comprehension, language production, speech recognition.

In the opinion of Pudasaini *et al.* (2022), recent developments in NLP have made it feasible to identify complex patterns seen in textual corpora. As a result, outcomes in various NLP tasks, including information extraction, text classification, POS tagging, and NER, were attained. The extraction of information has been applied to numerous business applications as well, since it involves removing informative texts from a given document.

As opined by Chowdhary (2020), the WWW contains at least 20 billion pages of data, making it a vast resource if critical information can be gleaned from them using natural language processing (NLP). This explains the reason about the importance of NLP. The following programs make it possible to make advantage of some of the most well-known applications of this data: in addition to searching and indexing huge documents, (IR) stands for information retrieval. The categorization of text into several departments, (IE) stands for information extraction. Text generating dialogues, automatic translation of languages, summarization of texts automatic, question-answering, and knowledge acquisition are all examples of automatic language translation.

2.3. NLP in Fraud Detection

A key aspect of protecting users and companies across a range of domains is fraud detection. The combination of NLP with text mining approaches improves the detection of fraudulent activity, especially when it comes to textual data like messages, emails, and posts on social media. There have many studies done under NLP to detect fraud in different domine.

In the research by M. Hamisu and A. Mansour (2020), a fraudulent email classifier (FEC) was created using natural language processing (NLP) to identify fraud. The classifier recognizes frequent terms and patterns used by scammers in their correspondence with victims by applying the Bag-of-Words (BoW) model. Text classification, information retrieval, and information extraction are all done with NLP approaches. The authors of this study gathered a dataset of fictitious emails from the Kaggle website and Nigeria's law enforcement agency. They created the Bag-of-Words as predictors and used machine learning algorithms for classifier to be train. Its purpose was to identify and categorize AFF emails coming from reliable sources. Through testing, the classifier indicated satisfactory performance in identifying and categorizing fake emails.

The research form Shahbazi and Byun (2021) state that by using NLP with ML (Machine Learning) algorithms it will be possible to evaluate and extract relevant information from text data which can further be used to detect fake news. They also explained how NPL contributes to the identification of language traits and patterns linked to false news, increasing the precision of fake news detection systems. Beside NLP blockchain technology was also utilized to secure the proposed solution and detect phony user accounts, while reinforcement learning was used to enhance the system's quality based on supplied information.

In the opinion of Lambo *et al.* (2022), the use NLP techniques that is text tokenization, stemming, and lemmatization as preliminary processing approaches facilitated the classification of texts for the identification and analysis social media posts related crime in their research. The study also employs a range of ML models, such as RF (random forest), SVM (support vector machine), KNN (K-nearest neighbour), and NB (naïve Bayesian) classifiers. According to the results, the SVM model has the highest accuracy, detecting crime in texts with an accuracy rate of 86%, while the RF model has an accuracy rate of 72% for criminal analysis. However, to increase the accuracy of crime detection, the researchers do admit the need for larger datasets and additional study must be taken. They also recommend that future research must be taken at the application of NLP (natural language processing) to internet slang and non-English languages and perform error analysis to learn more about the difficulties that NLP algorithms encounter in this attempt.

2.4. Text Mining

The process of digging through massive text archives to find hidden, interesting, irregular patterns or fresh information is called text mining. Finding relevant patterns in text documents to helps to locate the information needed which is a challenging task. Dang and Ahmad (2014) stated that vast amount of unstructured information spread across several locations which results as text mining becoming a significant area of study. It is estimated that unstructured text makes up 80%

of global data. Text mining is a field of study that includes machine learning, text analysis, information extraction, classification, clustering, visualization, and data mining.

In the opinion of Dang and Ahmad (2014), text mining algorithms can save time and money by providing us with structured, valuable data. Finding hidden information by text mining in the fields of biology, internet security, social media sites, etc. is an important task. The categorizing of text documents that contain links or other information is of great application due to the development of web technology.

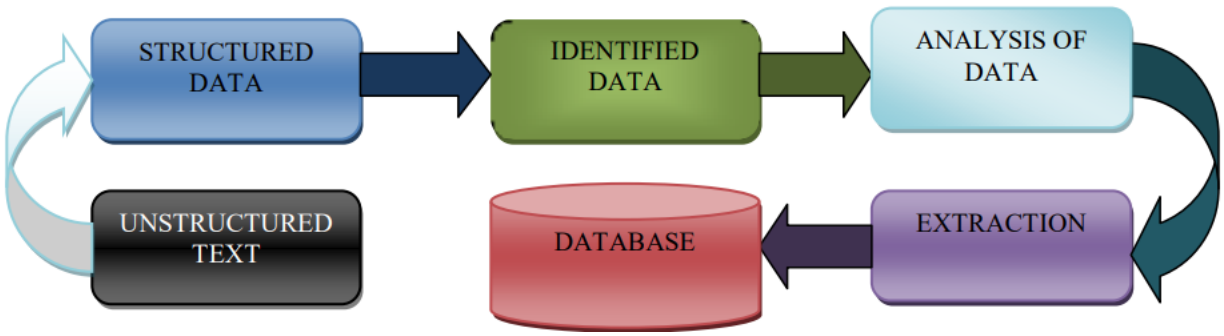


Figure 1: Processing of Text Mining (Dang and Ahmad, 2014)

2.5. Text Mining: An approach of Fraud Detection

Text mining is an effective technique for identifying fraud in a variety of contexts, including online conversations and financial transactions. It can be used to find anomalies and hidden patterns in data consist of text. For example, text mining can be used to examine financial data to spot odd language usage or ambiguous remarks that can point to fraud. In the same way, it can examine correspondence from customers, including emails and posts on social media, to spot unusual conduct or clues about potential fraud. Text mining can also be used to look at internet transactions, such as chat logs and product reviews, to identify fraudulent activity. Additionally, by examining the language used during social networking frauds, such as phishing emails and fraudulent websites, it is possible to identify trends in such communications. Thus, research in the subject of text mining is critical to creating more effective and potent techniques for drawing conclusions from textual data, which is necessary for many applications such as topic modelling, sentiment analysis, and fraud detection.

The researchers Yao, Zhang and Wang (2018), compares 120 fraudulent financial statements released by the China Securities Regulatory Commission based on size and industry, to show the value of data mining techniques in the identification of financial statement fraud. They used two feature selection techniques, PCA and Xgboost, to reduce dimensionality and five classification techniques, SVM, RF, DT, ANN, and LR, to construct the fraud detection models. The researchers proposed an optimized financial fraud recognition model to highlight the significance of data

mining approaches. This investigation demonstrated the effectiveness of data mining techniques in identifying financial statement fraud.

Sundar Kumar and Ravi (2013) proposed a text and data mining technique for malware detection. They used a dataset with several API calls and the static analysis method to find malware in the banking sector. Specifically, the API call sequences are the features that are extracted from the dataset using the text mining technique. In order to choose the qualities that are most pertinent for malware detection, the authors use mutual information. The dataset is balanced using over-sampling. After balancing the dataset, the study indicates that SVM and OCSVM reach 100% sensitivity in malware detection. Thus, the optimum method for detecting malware in the banking sector is thought to be static analysis with API calls. In addition, the writers examine pertinent literature and give a summary of data mining, feature extraction, and text mining techniques utilized in malware detection.

The solution that Sharmin and Zaman (2017) has designed to detect spam comments on YouTube videos specifically looks at the dangers and hazards that are presented by malevolent people which also utilize text mining to detect spam, phishing links, adverts, and false information in the comments section. The study classifies comments as authentic or spam using machine learning techniques and the text mining tool WEKA. For this, a few classification methods were also used, such as Support Vector Machine, Bagging, K Nearest Neighbour, and Naive Bayes. The paper also emphasizes how ensemble classifiers perform better in text classification than single classifiers. The article describes the technique, experimental setup, and acquired findings using the YouTube-Spam-Collection dataset obtained from the UCI machine learning data repository.

2.6. Related Work on social media Platform with different types of fraud detection

2.6.1. Cyberbullying Detection:

As compared to the traditional bullying notion, cyberbullying has more potent and long-lasting impacts because it can immediately reach many people. Furthermore, it may be difficult or time-consuming to remove online documents that contain such hazardous content in some circumstances. Online bullying may result in psychological illnesses like melancholy, low self-esteem, difficulty focusing, and even suicidal thoughts, even while it does not appear to inflict any immediate physical harm to the victim (Hosseinmardi et al., 2014).

Machine learning techniques combined with social media elements allow for the detection of cyberbullying. We can learn more about the trends and features of cyberbullying by examining the connection between social media traits and cyberbullying. Yin *et al*. (2009) provided the

results of the first study done for the automatic detection of cyberbullying. The study employed a linear kernel classifier in conjunction with feature extraction approaches (N-grams and Term Frequency-Inverse Term Frequency (TF-IDF)) on three distinct data sets to identify instances of harassment on online platforms. While the other datasets were from discussion-style forums like MySpace and Slashdot, one dataset was taken from a chat-style site called Kongregate. Even though the experimental results are insufficient, the work inspired more investigation.

The research in similar field presented by Bozyiğit, Utku and Nasibov (2021) created a balanced dataset of tagged materials with different social media properties. These characteristics included details on the sender's location, favorites, retweets, and following count. Through the application of the Chi-square test to the dataset analysis, it was possible to discern a high link between cyberbullying incidents and specific social media properties, like the quantity of followers. For instance, it was discovered that individuals who had more followers were less prone to abuse others online.

Balakrishnan *et al.* (2019) performs the research on cyberbullying detection on twitter. It uses Random Forest, a ML technique that generates numerous decision trees, each of which provides a distinct classifier, the cyberbullying was detected. The efficacy of the model was assessed by a series of tests carried out on data that was scraped from Twitter using the hash tag #GamerGate. The data included tweets along with additional metadata, like creation time, followers, and friends.

Choi, Jeon and Kim (2021) introduced a text mining and social network analysis approach for identification of key cyberbullies in social media environment. In order to identify prominent cyberbullies, researchers choose to identify the writers of hurtful comments. They classified major cyberbullies based on two characteristics which are individuals who make a lot of hurtful comments and have a big influence on the online community. During the analysis phase, the researchers calculated the cyberbullying index, employed the Losada ratio, and performed Social Network Analysis (SNA). They identified the salient traits of notable cyberbullies by analyzing textual data Cyberbullying Index, sentimental features, and social features (centrality indices). To calculate the cyberbullying score, the variables obtained from the analytical findings were multiplied together. This score serves as a valuable tool for identifying significant cyberbullies.

2.6.2. Phishing Detection:

Phishing is a serious risk on social media platforms since it is a fraudulent technique used to fool people into disclosing personal information. Phishing attacks become more likely to occur when consumers participate in different activities and give personal information. It becomes essential

to identify phishing attempts to preserve the security and integrity of social media platforms. Phishers must engage in extensive social engineering to plan and carry out these attacks, making their snares more reliable and focused on achieving certain goals (Kaivanto, 2014).

In the paper (Gouse Baig et al., 2020), ANN (artificial neural networks) and fuzzy inference systems are used to create an integrated ML (machine learning) framework that detects phishing. The model successfully classifies emails as legitimate or phishing by analysing phishing URLs (obfuscated, malicious, or phished URLs). To produce precise classifications, the model makes use of social aspects associated with phishing and features present in the URLs. ANN's Levenberg-Marquart and Feed-Forward Backpropagation techniques improve the URL classification process. In order to deal with imprecise data on social traits, the model also includes a fuzzy inference system. In an enterprise setting, this integrated design facilitates the possible to accurately detect and identify phishing emails, both known and unfamiliar.

Another research in similar field by (Zamir et al., 2020) used a stacking model in a supervised machine learning (ML) technique. In order to effectively detect phishing websites and identify phishing attempts, it combines lexical, URL-based, and feature selection techniques with machine learning algorithms. In order to conduct an analysis of the characteristics of the phishing data set, they used many techniques, such as information gain, gain ratio, Relief-F, and recursive feature removal. Following this, they use methods like as deep learning, SVM, and neural networks to the characteristics that they have selected, and then evaluate the outcomes by using measures such as accuracy, f-measure, recall, and precision. The stacking model obtains the best accuracy of 97.4% in identifying phishing websites by combining the classifiers with the highest scores.

Ozgur Koray et al. (2019) employed machine learning-based classification algorithms as a phishing detection technique. The study uses seven distinct machine learning algorithms to identify phishing websites: Adaboost, K-star, Decision Tree, (RF) Random Forest, SMO, and Naive Bayes. The writer states that identification process is based on examining web page URLs and utilizing a variety of feature sets, such as Word Vectors, NLP-based features, and Hybrid features. The study's dataset includes both authentic and phony web page URLs. 36,400 authentic URLs and 37,175 phishing URLs are totalled in the dataset. In order to identify new websites, the suggested approach additionally makes use of a sizable database of trustworthy and fraudulent websites.

2.6.3. Fake Reviews Detection:

The detection and mitigation of false or deceptive reviews on internet platforms constitutes the practice of fake review detection. It effects in customer trust and decision-making are greatly

influenced by user evaluations in the domains of e-commerce, hotel, and other online services. However, a major obstacle to preserving the integrity of online platforms is the prevalence of fake reviews, which are frequently created intentionally. The detection of fake reviews is a multifaceted strategy that combines technology advancements, user education, and platform collaboration to foster an atmosphere in which real user opinion is dominant.

Numerous studies have been carried out to lessen the number of fake reviews. A feature structure that divided reviews into two categories was used in the study (Barbado et al., 2019) to create a phony review detection system. The F-score indicates that 86% of the user-centric elements in Amazon electronic product reviews were accurately analysed using logistic regression. The method made use of Opinion Mining and Sentiment Analysis to find reviews that were fraudulent. The sentiment polarity of the written passages was ascertained by using Vader's Sentiment Analysis to the annotations that the researchers had gathered for each review in the data set (positive, negative, or neutral). Most sentiment analysis techniques can be classified as valence-based or polarity-based, depending on how strongly a sentiment is considered. For instance, "excellent" is viewed as more positive than "good" in a valence-based approach, but in a polarity-based approach, they are rated similarly.

In another study (Elmogy et al., 2021) concentrated on two primary categories of fraudulent reviews: textual (based on the reviews' content) and behavioral (based on the reviewer's writing style, emotional emotions, and frequency of writing). Algorithms for machine learning were used to distinguish between real and fraudulent reviews. In the identification procedure, "important elements of the review" and "reviewer behavior" were considered. Without taking behavioral traits into account, Logistic Regression obtained an accuracy of 86% in bi-gram analysis, KNN 73% in tri-gram analysis, and SVM 88.1% in bi-gram analysis.

The authors of (Elmurngi and Gherbi, 2017) employed supervised machine learning techniques to detect fraudulent reviews in one of their research projects. There are five classifiers used: SVM, KNN, k-star, Naive-bayes, and decision tree. The labeled movie reviews dataset has been subjected to three types of simulation tests, with 1400, 2000, and 10662 movie reviews in each edition. Additionally, the authors employed maximum entropy, Naive Bayes, Decision trees, SVM, Random forests, and Random forests as classifiers to identify fraudulent reviews on the dataset they had gathered. The gathered information consists of approximately 10,000 unfavorable tweets about Samsung goods and services.

Table 1: Summarization of Literature Review

Reference	Platform/Domin	Data Used	Type of Fraud	Techniques Used
M. Hamisu and A. Mansour (2020)	Cybercrimes domain	Work leveraged 4500 fraudulent emails	Advance Fee Fraud (AFF)	• Bag-of-Word mode • Natural Language Processing • Decision Tree • Discriminant Analysis • Logic Regression • Support Vector Machine • Nearest Neighbor Ensemble
Shahbazi and Byun (2021)	Fake News Detection	Twitter, Facebook, BBC news, etc. data contains 900.000 records 2015-2019.	Misinformation or Rumor	• Natural Language Processing • Blockchain Technology
Lombo, Oyelade, Olaide N and Ezugwu, Absalom E (2022)	Crime Detection	Twitter Spam dataset	Law enforcement and criminal justice	• Natural Language Learning • TF-IDF (Term Frequency- • Inverse Document Frequency) • SVM algorithms • Naive Bayesian algorithms • KNN classification algorithms
Yao, Zhang and Wang (2018)	Financial Statement Fraud Detection	120 fraudulent financial statements by China Securities Regulatory Commission	Financial Fraud	• Text Mining • PCA and Xgboost • SVM • RF • DT • ANN • LR
Sundarkumar and Ravi (2013)	Windows Operating System	dataset with several API calls	Malware Detection	• Text Mining • SVM (OCSVM) • Decision Tree (DT) • Probabilistic Neural Network (PNN) • Multi-Layer Perceptron (MLP) • Group Method of Data Handling (GMDH)
Sharmin and Zaman (2017)	Online social media	YouTube spam dataset from UCI-machine learning data repositor	Spam Detection	• Text Mining • Naive Bayes • K Nearest Neighbor (KNN) • Bagging • Support Vector Machine
Yin *et al.* (2009)	Wed 2.0	Fundaci´on Barcelona Media (FBM) datasets: Kongregate, Slashdot and MySpace	Cyberbullying Detection	• TFIDF with Sentiment and Contextual Features • N-grams as binary feature
Bozyiğit, Utku and Nasibov (2021)	Social media features	Twitter Data from API (124317 Tweets)	Cyberbullying Detection	• Term Frequency – In-verse Document Frequency (TF IDF) • Support vector machines (SVM) • Logistic regression (LR) • K-nearest neighbor (KNN) • Naive bayes multinomial (NBM) • Adaptive boosting (AdaBoost • Random forest (RF
Balakrishnan *et al.* (2019)	Twitter	data crawled. from Twitter, using #GamerGate	Cyberbullying Detection	• Big Five and Dark Triad features. • Random Forest technique
Choi, Jeon and Kim (2021)	Daum Agora online discussion community	Crawled 650000 posts and comments from online community	Cyberbullying Detection	• Text mining • social network analysis approach • Random Forest • Logistic Regression • SVM
Gouse Baig et al., (2020)	E-mail Phishing Detection	online data of the 2000 phishing URLs and legitimate	Phishing Attack	• ANN (artificial neural networks) • fuzzy inference systems • Neural Network for analysis

Zamir et al., (2020)	Phishing web site detection	freely available datasets from kaggel.com (11,055 websites, and 32 attribute)	Phishing Attack	• Information gain • Relief-F • Principal components analysis • Recursive feature elimination • deep learning • support vector machines • neural networks
Ozgur Koray et al. (2019)	Phishing detection from URLs	dataset with 36,400 legitimate URLs and 37,175 phishing URLs from Yandex Search API	Phishing Attack	• Adaboost, • K-star • Decision Tree • (RF) Random Forest • SMO • Naive Bayes • NLP-based Hybrid features
Barbado et al., (2019)	Online consumer electronics retailers	Scraping with help of Yelp on electronics reviews	Fake reviews	• LR (Logistic Regression) • DT (Decision Tree) • RF (Random Forest) • GNB (Gaussian Naive Baye) • AB (AdaBoost))
Elmogy et al. (2021)	Hotels Reviews	Yelp dataset includes 5853 reviews of 201 hotels in Chicago. written by 38, 063 reviewers.	Fake reviews	• K-Nearest Neighbors algorithm • Decision-tree • Random Forest • Logistic regression • Support Vector Machines (SVM) • Naive Bayes (NB)
Elmurngi and Gherbi (2017)	Movies Reviews	movie reviews 1400, 2000, and 10662 movie reviews	Fake reviews	• SVM • KNN • k-star • Naive-bayes • decision tree

2.7. Literature Gap

The body of research on using NLP and text mining to detect fraud in social media has thoroughly examined a range of methods and strategies for locating and preventing fraudulent activity. A significant gap in the existing body of literature is not having of comprehensive qualitative and quantitative assessments about the efficacy of NLP and Text Mining models designed with social media fraud detection in mind. Although a large body of research has been done on the use of these technologies for fraud detection, very few of these studies have included comprehensive quantitative metrics to assess the performance and accuracy of these models in real-world social media situations.

This study provides a detailed numerical evaluation of the efficiency of text mining and natural language processing approaches for social media fraud detection in this research article. The purpose of this research is to fill a vacuum in the existing body of knowledge. The evaluation of significant metrics such as recall, precision, and F1-score will be a part of this process. The goal is to provide a quantitative picture of the effectiveness of the offered methodologies in detecting and intercepting social media fraud.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1. Introduction

Research methodology refers to the procedure utilized to carry out a research investigation. This covers all plans, procedures, strategies, and techniques applied to the gathering, evaluating, and interpreting of data (Synder, 2019). A machine learning approach, which is a subset of data science, has been employed to conduct this study. These multiple disciplines area gathers, processes, and analyses data to learn new things or find solutions to problems (Glusker, 2023). It does this by combining programming, computer science, statistics, and domain expertise. There are different types of ML methodologies for conducting a studies on ML, such as:

Supervised learning: The method of learning is known as supervised learning which involves optimizing the minimized parameters to the difference between the computed output and the intended outcome (Jo, 2021). The supervised learning framework includes the KNN, Naive Bayes, and decision tree algorithms. These learning algorithms are applied in the domains of regression and classification.

Semi-supervised learning: According to Bewtra (2022), one category of supervised learning tasks and methods that uses unlabelled data for training is called semi-supervised learning. A substantial number of unlabelled data is usually accompanied by a modest amount of annotated data. The two types of learning supervised and unsupervised are represented by semi-supervised learning, which is the latter having fully labelled training data and the former lacking any.

Unsupervised learning: The technique of optimizing cluster prototypes based on similarities between training samples is known as unsupervised learning (Jo, 2021). The goal that the unsupervised learning algorithms are used to is clustering data elements.

Reinforcement learning: Reinforcement learning describes strategies for increasing performance through experience (Barto, 1997). Even with the latest advancements in artificial learning systems such as improved techniques for training artificial neural networks the majority of these systems still pick up knowledge from an experienced “teacher” who can instruct them on how to react to a given set of training stimuli.

3.2. Research Approach

During this study, the method considered was supervised machine learning ideas for the model implementation step. A effectives method for teaching computers to learn from labelled data and forecast new information is supervised learning. The main objective is to develop models that may be utilized in real-world situations by employing techniques like NLP with text mining utilizing Naive Bayes algorithm combined to N-grams and TF-IDF feature extraction concept. Translating theoretical knowledge into real-world applications where data and algorithms combine to yield insightful outcomes is focused on this chapter. Numerous domains, including healthcare (Uddin et al., 2019), finance (Tsantekidis et al., 2017), and NLP (Shankar et al., 2022), have demonstrated the impact of supervised learning through studies. This experiential stage is essential to transforming research ideas into practical.

3.3. Model Implementation

This article explores social media fraud detection, an intriguing issue in current social media platforms. We will study a feature extraction approach that combines N-grams with TF-IDF which are powerful techniques for extraction of key data items needed to identify social media fraud. Naïve Bayes will be our selected method for model selection and training.

3.3.1. NLP in Model

Data cleaning and pre-processing were conducted after data collection to enhance the accuracy of the subsequent analysis and prediction performed by the models. The first step was removing special letters and numbers from the text, followed by deconstruction, which entails transforming shortened and associated words into their whole form. Word segmentation (tokenization) and lemmatization were performed to transform the words into their standardized form. Stemming was used with lemmatization due to its capacity to provide very favourable results, even in the absence of a lexicon. Employing the stop word removal approach was crucial because of its substantial influence on the precision of deception prediction. The combination technique was chosen due to its improved prediction accuracy after examining the performance of the models using TF-IDF vectorization and N-gram method.

It was decided to choose Naive Bayes model out of many machine learning models. The model was developed, evaluated, and tested using aggregated data to compare different natural language processing techniques and provide insight into the models' potential for mutual application and generalization. The model underwent testing using a broader spectrum and mix of n-grams, as well as a varied number of characteristics, to choose the models that exhibited the best level of

prediction accuracy. A comparison was made between models that used n-grams, TF-IDF vectorization, and a mixture of both techniques. During the procedure, the most crucial characteristics were chosen using the Python 3.12.0 environment.

3.3.2. Extracting Features:

Feature extraction improves the performance of pattern recognition or machine learning. As per Lee and Landgrebe (2019), feature extraction is a process that simplifies data by focusing on its fundamental elements, which is important for machines and deep learning models. The strategy involves eliminating unnecessary characteristics from the data, possibly resulting in a decrease in model accuracy. Text categorization, including social media communications, requires vector representation of documents by extracting features. In social media fraud detection analysis, features are selected and represented using NLP, Bow (Bag of Words), TF-IDF, and N-grams among which we are being using (TF-IDF) and N-grams in this research.

3.3.2.1. N-grams Features Extraction:

The feature extraction method, N-grams enable the examination of contiguous word groupings to capture the context and associations within the text. The extraction of N-gram features is a highly potent technique within the realm of natural language processing (NLP). This methodology revolves around the acquisition of sequential groupings of neighbouring words, commonly referred to as "n-grams," from a provided textual corpus. N-grams offer significant insights into the localized context and textual patterns, facilitating the discernment of meaningful correlations among words. The variable denoted by the letter "N" in the term "N-gram" signifies the cardinality of the word sequence, specifically referring to the number of words encompassed within each individual sequence.

In the given sentence "Everything is good so far." the inclusion of bigram (2-gram) features can be observed. These features consist of “Everything is”, “is good", “good so” and “so far” The utilization of trigrams, also known as 3-grams, allows for a more granular dissection of the given sequence, resulting in the identification of two distinct components: " Everything is good ", " is good so" and “good so far”.

The utilization of N-Gram feature extraction has been observed to yield significant advantages in various domains, including but not limited to sentiment analysis, language modelling, and even in fraud detection. N-grams play a significant role in capturing the intricate structure and nuances of language by considering the sequential relationships between words. Since it helps in interpreting the context of a certain text, this is of extreme significance.

Within the framework this research has employed the method of N-Gram feature extraction in conjunction with other methodologies, such as TF-IDF, to augment the representation of textual data. The integration of this combined approach facilitates a more exhaustive examination, encompassing both localized and global patterns within the textual data. As a result, it enhances the efficacy of the machine learning model.

3.3.2.2. TF-IDF Features Extraction:

This feature extraction tool is used to degrade words that occur often but have little significance within the document's context (Tabassum and Patil, 2020). Term Frequency denoted by (TF) and Inverse Document Frequency denoted by (IDF) are the two primary components that have been included into the notion. (TF) can detect the frequency of a word in a text without the size of the document being affected in any manner. This is possible by using the word frequency. It is possible for it to deliver a degree of accuracy that is equivalent to both small and big datasets since it is classified by the length of the document as a approach of normalization. This is the reason why this is the case. It is possible to determine the relevance of a word in the paper by calculating its inverse document frequency, often known as IDF. Essentially, the volume of words that are of less relevance may be reduced by computing the logarithm of the relationship of the total amount of documents to the total number of occurrences the word occurs in that document. This brings to a reduction in the overall number of words. This is accomplished by reducing the number of documents that include the word. Equation 1 shows the weight of a term calculation by multiplying two frequency types.

$$TF - IDF = TF * \log(\frac{N}{DF}) \qquad (1)$$

This extraction tool is used in this study for the purpose of extracting the characteristics of the contents of two different models, which are bi-gram and trigram. Additionally, the expanded dataset is used in both language models once the characteristics that are representative of the actions of the users have been extracted.

3.3.3. Selection and Training of the Model: Naïve Bayes:

After extracting the characteristics, it is important for to choose a appropriate model for fraud detection. The selection and training of an appropriate model are crucial in machine learning. These processes significantly impact system performance and outcomes. The Naïve Bayes method is a popular and effective classification approach. It is particularly known for its simplicity, speed, and resilience for classification. In this part, we go into the reasoning for

selecting Naïve Bayes, examine its inner workings, and discuss the stages required in the training process of this algorithm. Naïve Bayes, an algorithm that is both simple and powerful, is an ideal choice. Naïve Bayes uses probability theory to categorize text into distinct classifications. Even if it uses labelled data, the model undergoes training to acquire knowledge about the distinctive patterns linked to both authentic and deceitful information. This enables it to provide precise forecasts when encountering novel, unobserved data.

3.3.3.1. Naïve Bayes: Theoretical Approach

The Naïve Bayes classification method is a technique for classification that adopted from Bayes theorem, as stated by Lindley in 1958. When considering the information that is currently available regarding the conditions that are associated with a particular occurrence, this theorem can be utilized to quantify the likelihood of that event taking place. This classifier assumes that a particular characteristic within a class does not have a direct correlation with any other characteristic, even if the characteristics that are contained within that class may have interdependencies with one another (Rish, 2001).

Figure 2 shows how the Naive Bayes approach classifies a new item (white circle) as 'green' or ‘red'. This image suggests that a new item's 'green' membership is twice as likely as ‘red' membership. This is because 'green' objects (40) outnumber ‘red' objects. In Bayesian analysis, prior probability exists. The initial probabilities for 'green' and ‘red' are 0.67 (40 × 60) and 0.33 (20 ÷ 60), respectively. We must enclose the 'white' object in a circular boundary with many selected points regardless of class labels to categorize it. Three ‘red' and one 'green' points made up this figure. Thus, the probability of 'white' given 'green' is 0.025 (1 divided by 40) and given ‘red' is 0.15 (3 divided by 20). Although the initial probability suggests the new item is 'green', the likelihood suggests it is ‘red'. Bayesian analysis creates the classifier from previous probability and likelihood values. These two sources of information are combined using the ‘multiplication' function to create the 'posterior' probability product. The posterior probability of 'white' being 'green' is 0.017, calculated by multiplying the prior probability (0.67) by the conditional chance of 'green' given 'white' (0.025). The subsequent probability of 'white' being ‘red' is 0.049, calculated by multiplying the prior probability of 'white' (0.33) by the conditional probability of ‘red' given 'white' (0.15). According to Naive Bayes, the new white item should be classified as red.

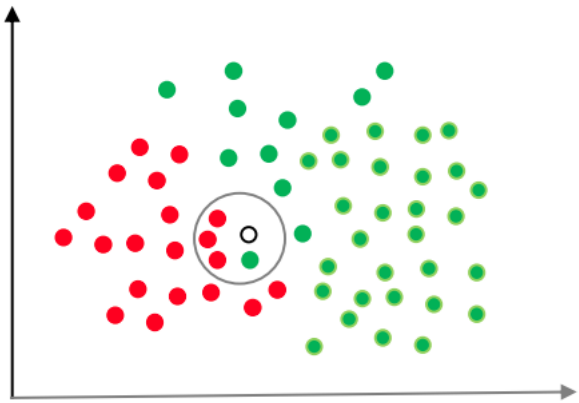


Figure 2: Naïve Bayes algorithm illustration.

Bayesian, or generative, classifiers model word characteristics in distinct classes to create a probabilistic classifier, according to Aggarwal et al. (2012). Text is categorized by calculating the posterior probability of documents belonging to different classes based on word occurrence. Bayes' theorem for a document d and a class c may be stated as follows:

$$P(c|\mathrm{d}) = \frac{P(d|\mathrm{c}).P(c)}{P(d)} \qquad (2)$$

Equation (2) represents calculation of class posterior probability, denoted as $P(c|d)$. It is determined by the probability of the class, $P(c)$ and the probable outcome of the predictor given the class $P(d|c)$.

It has been stated by Raschka (2014) that Naive Baye's decision rule classifier that is based on the posterior probability may be phrased as follows in the context of fraud classification:

$$if\ P(c = fraud|\mathrm{d}) \geq P(c = no_fraud|\mathrm{d})\ \text{classify as fraud.}$$

$$\text{else classify as no_fraud} \qquad (3)$$

The following probability is the sum of the class-conditional probability and the initial probability. If the evidence element is constant for both classes, it can be omitted.

$$P(c = fraud|\mathrm{d}) = P(d|\mathrm{c} = \mathrm{fraud}).P(fraud)$$

$$P(c = no_fraud|\mathrm{d}) = P(d|\mathrm{c} = \mathrm{no_fraud}).P(fraud) \qquad (4)$$

In order to get the prior probability, one may use the maximum likelihood estimate, which is derived from the frequencies of fraud and non-fraud tweets that are included in the training dataset.

$$\hat{P}(c = fraud) = \frac{\#\ of\ fraud\ tweets}{\#\ of\ total\ tweets}$$

$$\hat{P}(c = not_fraud) = \frac{\#\ of\ not_fraud\ tweets}{\#\ of\ total\ tweets} \quad (5)$$

Two discrete models are available for the computation of class-conditional probabilities. The Multinomial model and the Multivariate Bernoulli model comprise these models. A naive assumption has been made regarding the conditional independence of the words in each document.

According to Murphy's (2012) explanation in one of his articles, document categorization refers to the challenge of dividing text documents into a variety of categories. One simple method is to represent each document as a binary vector that records whether each word is present. Let x_i represent a vector of counts for document i, such that $x_{ij} \in \{0, 1,..., N_i\}$, where N_i is the number of words in document i $\left(i.e \sum_{j=1}^{D} x_{i,j} = N_i\right)$. The multinomial distribution listed below may be applied to the class conditional concentrations:

$$p(x_i \mid y_i = c, \theta) = Mu\left(x_{ij} \middle| N_i, \theta_c\right) = \frac{N_i!}{\prod_{j=1}^{D} x_{ij}!} \prod_{j=1}^{D} \theta_{jc}^{x_{ij}} \quad (6)$$

3.4. Model Evaluation

In machine learning, model evaluation is very important component of the resulting any process, offering useful insights into the capabilities and limitations of our models, and ensuring their suitability for the intended use. We may optimize our models' efficacy and make well-informed judgments on their deployment by thoroughly assessing them and choosing the best ones.

3.4.1. Confusion matrix:

A tabular representation that displays the overall output of a classification issue, including both the anticipated and observed values generated by a classifier is called confusion matrix. It is used to visually represent the results of the prediction and aids in understanding the performance of the classifiers. The confusion matrix consists of four distinct sorts of values which are shown in figure 3.

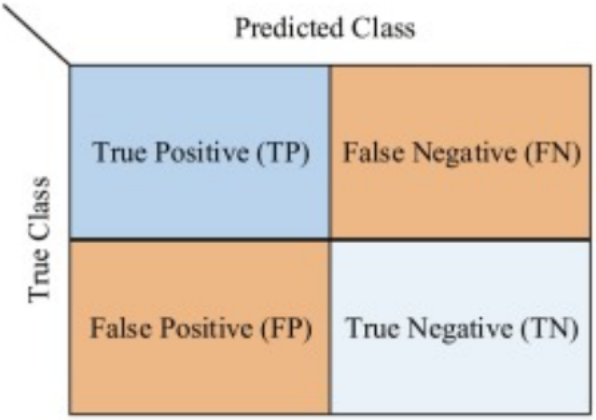


Figure 3: Confusion matrix

3.4.2. Accuracy:

The ratio of properly categorized data cases to the total number of data cases, providing insight into the number of data instances that were accurately classified is referred to a measure of accuracy.

$$Accuracy = \frac{TP+TN}{TP+TN+FN+FP} \quad (7)$$

3.4.3. Precision:

The accuracy with which a classifier generates positive predictions is measured. This phrase's definition is the proportion of correctly predicted positive outcomes to all positive predictions made by the classifier.

$$Precision = \frac{TP}{TP+FP} \quad (8)$$

3.4.4. Recall:

The percentage of positive instances in the data that the classifier correctly predicted out of the total number of positive cases is recall.

$$Recall = \frac{TP}{FN+TP} \quad (9)$$

3.4.5. F1-Score:

A statistic known as the F1-Score is a combined approach of a classifier's precision and recall that is calculated. The HM (Harmonic Mean) of recall and precision is the approach that is used to calculate it. This method is distinct from the way of calculating the "average" of the variables. When it comes to ratios such as accuracy and memory, the conventional arithmetic means is less suited than the harmonic mean.

$$F1 - Score = 2 * \frac{Precision*Recall}{Precision+Recall} \quad (10)$$

CHAPTER FOUR: DATA

A discussion of the dataset that we will be employing for this project has also been included in this chapter, which offers information on the datasets that are currently accessible. Following this, there will be a discussion on the data analysis that is carried out on our dataset, in addition to the steps of data preparation, all of which are covered in more depth in this chapter.

4.1. Data Collection

In order to perform experimental analysis three distinct Twitter datasets with attention to identify fraud for the practical inquiry for this research work. Every dataset concentrates on a certain scam type that is now well-known on social media. These datasets were specifically chosen to represent a broad spectrum of online fraud. The information was obtained from the trustworthy web resource kaggle.com. The databases contain examples of misinformation, fake reviews, and cyberbullying. The datasets were selected to obtain a wide range of fraudulent activities taking place on different social media platforms. Thus, including these diverse datasets, it allows to able to further investigate the various obstacles to fraud detection in our research and get insight into the intricate realm of online fraud.

Table 2: Summary of Data Collected from Kaggel.com

Source	Fraud Type	Number of Data	Labeled
	Cyberbullying	46,017	Yes (5 Category)
	Fake Reviews	40,000	Yes (2 Category)
	Misinformation	10,877	Yes (2 Category)

The first dataset is composed of cyberbullying tweets in two different columns. It consists of 46,017 unique values of tweet within tweet text, and another is cyberbullying type with different bullying types for example bullying related to culture, gender, age, ethnicity, and other cyberbullying (Wang and Lu, 2020). Likewise, the second dataset is about reviews which are fake containing twenty thousand fake reviews and twenty thousand real product reviews. This dataset was created and used in research by Salminen et al. (2022), for the model created to detect fake reviews in online platforms. It consists of OR-Original and CG (Computer-generated fake reviews). Finally, the third dataset is about the collection of tweets about disasters. The dataset

aims to facilitate the identification of relevant information about disasters from social media. It consists of tweets that have been manually classified into two categories: relevant and irrelevant. Among 10,877 unique tweets it consists of 57% not relevant and the remaining 42% relevant in nature.

Table 3: Sample of Cyberbullying Tweets Dataset

	tweet_text	cyberbullying_type
0	In other words #katandandre, your food was cra...	not_cyberbullying
1	Why is #aussietv so white? #MKR #theblock #ImA...	not_cyberbullying
2	@XochitlSuckkks a classy whore? Or more red ve...	not_cyberbullying
3	@Jason_Gio meh. :P thanks for the heads up, b...	not_cyberbullying
4	@RudhoeEnglish This is an ISIS account pretend...	not_cyberbullying

Table 4: Sample of Fake Review Tweets Dataset

```
df_fakeReviews.head(5)
```

	category	rating	label	text_
0	Home_and_Kitchen_5	5.0	CG	Love this! Well made, sturdy, and very comfor...
1	Home_and_Kitchen_5	5.0	CG	love it, a great upgrade from the original. I...
2	Home_and_Kitchen_5	5.0	CG	This pillow saved my back. I love the look and...
3	Home_and_Kitchen_5	1.0	CG	Missing information on how to use it, but it i...
4	Home_and_Kitchen_5	5.0	CG	Very nice set. Good quality. We have had the s...

Table 5: Sample of Disaster Tweets Dataset

```
[18] df_disaster.head(5)
```

	tweetid	text	Labeled
0	1.0	Just happened a terrible car crash	Relevant
1	13.0	Our Deeds are the Reason of this #earthquake M...	Relevant
2	14.0	Heard about #earthquake is different cities, s...	Relevant
3	15.0	there is a forest fire at spot pond, geese are...	Relevant
4	16.0	Forest fire near La Ronge Sask. Canada	Relevant

4.2. Data Pre-Processing

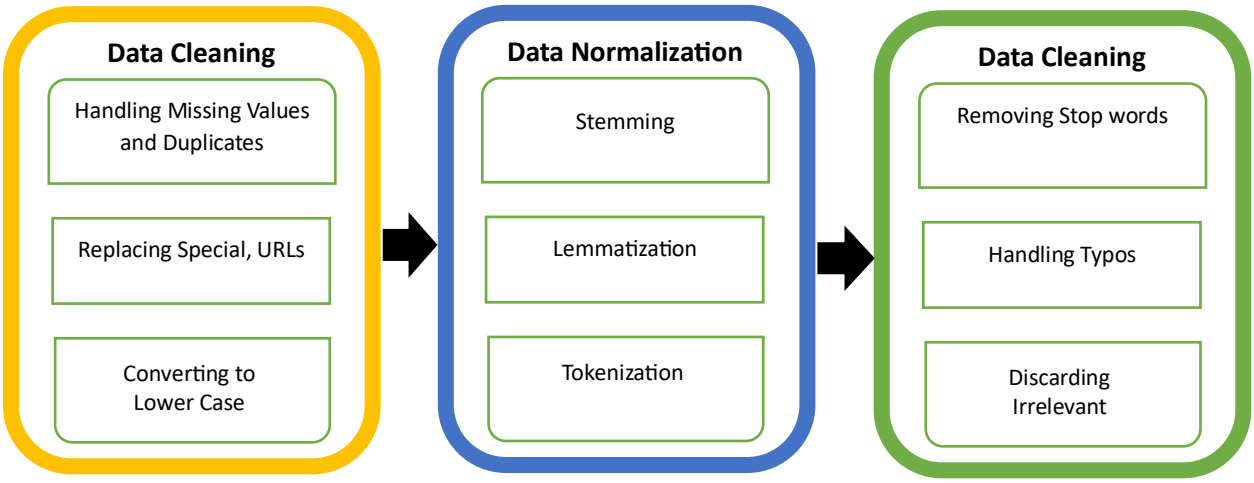


Figure 4: Data Pre-Processing Steps

The text was cleaned using Natural Language Learning algorithms. Text message data lacks text data, such as numbers and emojis, and is mostly noise. The different type of emojis will be converted to text, all characters will be converted to lowercase, numbers will be converted to words, punctuations will be removed, mentions, white spaces, stop words, and URLs will be changed replaced as part of the data cleaning process. Tokenization, stemming, and word lemmatization are NLP techniques that are used to break down words into their most fundamental form. The data pre-processing overview is displayed in Figure 1.

4.3. Using NLP in Data Pre-processing

Natural Language Processing (NLP) is an important tool in data preparation field for my research study, particularly when dealing with textual data. Natural Language Processing (NLP) significantly aids in the data refining and structuring for machine learning applications and subsequent analysis by means of the utilization of diverse methodologies. Tokenization involves breaking down text into meaningful units, such as words or phrases. For example, the line "Hello, how are you?" may be converted into separate tokens: "'Hello', ',', 'how', 'are', 'you', '?'". Lowercasing standardizes case sensitivity by transforming all words to lowercase, assuring uniformity. The act of removing punctuation involves excluding unnecessary symbols, which enhances the readability and uniformity of the text. Removing stop words aims to remove frequent, insignificant words like "the" and "and," in order to preserve only contextually relevant keywords.

Furthermore, stemming involves reducing words to their fundamental form, shown by the conversion of "running" to "run," whereas lemmatization is the process of combining words based

on their dictionary meaning or base form, as demonstrated by the lemmatization of "better" to "good." The use of spell-checking software corrects typographical mistakes, particularly those in which "hte" is misspelled as "the." A more comprehensive approach is required to fix typographical problems, and this includes replacing "teh" as "the." In addition, steps are taken to remove HTML components and replace URLs with placeholders like "URL," ensuring a cleaner dataset. Tokenizing words and phrases make material easier to handle and allows for more in-depth analysis. Furthermore, part-of-speech labelling enhances linguistic understanding by assigning labels to grammatical categories, such as nouns and verbs. Named Entity Recognition (NER) is an illustrative method that involves recognizing and classifying entities. For instance, NER may recognize "New York" as a place and "Apple" as a corporation. Each NLP approach contributes to refining textual data, guaranteeing its alignment with the parameters of the study dataset. This comprehensive technique serves as a solid foundation for later assessments and the implementation of machine learning models.

4.4. Data Cleaning

Text noise elimination has been accomplished by data cleaning process in this step. Data cleaning includes handling missing values and duplicates, replacing special characters, URLs and user mentions and finally converting letters to lower case.

Handling Missing Values and Duplicates:

In order to ensure that this analysis is based on complete and unique information, handling missing values and duplicates must have been considered to eliminate errors, and ensure the trustworthiness of the result.

Replacing Punctuations, Specials, URLs and User Mentions:

Punctuations, especially hashtags, are unnecessary because our attention focuses just on the text (words). For this reason, in this step, we eliminated all punctuation from the sentences.

All URLs are replaced or removed because we do not want to monitor any text that contains a url in the unlikely event that a model identifies a patten in them for fraud detection.

Every mention, like (@Jason_Gio), is replaced and is substituted so that the training model may identify patterns that characterize specific phrases uploaded or used by certain users. As we know the quantity, we may utilize the numbers to identify patterns in fraud, therefore this stage also entails translating all the numbers into words.

Example 1:

Input: Why is #aussietv so white? #MKR #theblock #ImACelebrityAU #today #sunrise #studio10 #Neighbours #WonderlandTen #etc

Output: Why is aussietv so white MKR theblock ImACelebrityAU today sunrise studio10 Neighbours WonderlandTen etc.

Example 2:

Input: Accident cleared in #PaTurnpike on PATP EB between PA-18 and Cranberry slow back to #traffic http://t.co/SL0Oqn0Vyr

Output: Accident cleared in PaTurnpike on PATP EB between PA18 and Cranberry slow back to traffic URL

Example 3:

Input: @Jason_Gio meh. :P thanks for the heads up, but not too concerned about another angry dude on twitter.

Output: User meh P thanks for the heads up but not too concerned about another angry dude on twitter

Removing the URL and the user mention from Examples 1 and 2 could cause the fraud pattern to disappear. While the numerical values are not that important or significant in determining the fraud detection, understanding the numerical value linked to specific phrases might aid in the detection of criminal activity, particularly in statements associated with criminal activity.

Converting to Lower Case:

In this step we also converted all the letters to lower case to make sure that words in different instances are handled consistently to encourage precise representation and interpretation of the text.

Example 1:

Input: Karma. I hope it bites Kat on the butt. She is just nasty. #mkr

Output: karma i hope it bites kat on the butt she is just nasty mkr

4.4. Data Normalization

This process standardizes the text data to reduce variations and facilitate feature extraction. This involves tasks like:

Tokenization*:*

The process of tokenization involves breaking up the input text, which is merely a lengthy string of characters, into smaller pieces that are referred to as tokens by a computer (Grefenstette, 1999). It describes the division of a text sequence into smaller units. These tokens can range in length from words to characters. This procedure is important primarily because it facilitates machine comprehension of human language by reducing it to smaller, more manageable chunks for analysis.

Tokenization divides sentences, words, letters, and subworlds. Almost every NLP pipeline starts with tokenization. It greatly affects other pipeline components. Tokenization divides unstructured data, including natural language text, into discrete information bits. A vector representation of a document may be done instantaneously using its token occurrences. An unstructured string (text document) is instantly turned into a numerical data structure appropriate for machine learning using this approach. Their direct usage by a computer may start useful actions and responses.

Example 1:

Input: `'Just happened a terrible car crash'`

Output: `['Just','happened','a','terrible','car','crash']`

Stemming*:*

In general, stemming is the text mining process which refers to an imprecise algorithmic procedure that, to accomplish this objective, the majority of the time, involves removing derivatives from words (Cambridge University Press, 2009). Porter's algorithm is the most popular algorithm for stemming English, and it has been consistently demonstrated to be extremely successful in empirical studies (Porter, 1980). The Porter Stemmer algorithm is quick, widely used, and mild, it was utilized for stemming in this research as well.

Table 6: Rules of porter stemmer

(F)	**Rule**			**Example**		
	SSES	→	SS	caresses	→	caress
	IES	→	I	ponies	→	poni
	SS	→	SS	caress	→	caress
	S	→		cats	→	cat

Lemmatization

According to Cambridge University Press (2009), lemmatization is defined as the process of correctly using lexical and grammatical evaluation of words with the goal of eliminating only inflectional endings and restoring the lemma, or dictionary form, of a word. This process is performed with the intention of removing only inflectional endings.

Lemmatization and stemming are similar processes. The aim in lemmatization and stemming is to break down a word to its most basic form. The root is known as a lemma during the lemmatization process and as a stem during the stemming process. In stemming, some of the word's end is cut off without thinking about what it means to get to the stem. In lemmatization, on the other hand, the algorithm knows what the word means and works like an algorithm that looks up the meaning of a term in a dictionary before fragmenting it down toward its lemma, or root word. For example, the word ‘went’ originated from the word ‘go’, the lemma of ‘went’ is ‘go’, according to the lemmatization method. For improved outcomes, we chose to apply both algorithms: lemmatization was applied after the stemming procedure (Table 7).

Table 7: Example of Lemmatized words.

Words	Lemma
running	run
better	good
happily	happy
went	go
playing	play
largest	large
worst	bad
cities	city
faster	fast
happiest	happy

Removing Stop words

A group of words that are often employed in a particular language, such as conjunctions, prepositions, or articles, are referred to as stop words from that language. The addition of these words to a statement does not contribute much to its meaning. By ignoring them, it is possible to avoid sacrificing the what the text is trying to say. The removal of English Stop Words occurred during this time. The removing of stop words is a crucial step in NLP. In the field of NLP, words that are not useful (data) are referred to as stop words. In the past, eliminating stop words was mostly done to speed up text mining computations. This technique, which can be thought of as a dimensionality reduction of text data, was frequently applied by search engines to get better results (Huston and Croft 2010). An example of a stop word may be seen in Figure 3. It is possible to alter this list by adding words of your choosing that are written in English.

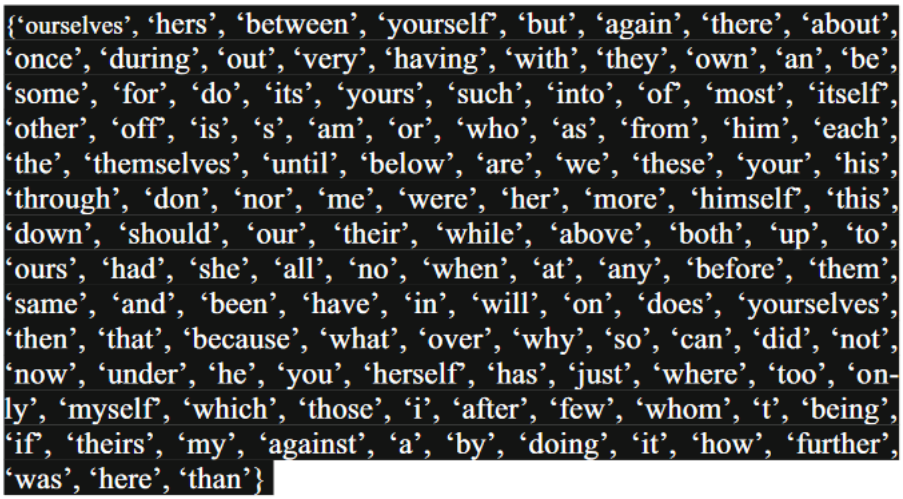


Figure 5: Some of the examples of Stop words.

Example 1:

Input: `['Just','happened','a','terrible','car','crash']`

Output: `[ 'happened','terrible','car','crash']`

Handling Typos

In the context of NLP or text mining, managing spelling also known as typos is crucial for enhancing the precision of search engines, language models, and other systems that use textual data. For the purpose of improving the quality of the data related to text, mistakes must be found and either corrected or mitigated. It has been used in this research because it is important dealing with jobs involving NLP, since the level of performance in the quantity of the input text can have a big effect on how well language models and algorithms work.

They are errors made when typing or writing that lead to misspelled words or inappropriate character sets. These mistakes can be made for several reasons, such as carelessness, inattention, or not knowing the proper spelling. There are several methods for dealing with typos. Spell checking: It is the process of comparing misspelled words to a dictionary of correctly spelled words to identify them and fix them using algorithms or other methods. Fuzzy Matching: Using algorithms to estimate string matching, which is helpful in locating near matches for words that have been misspelled.

CHAPTER FIVE: EXPERIMENTS AND RESULTS

5.1. Experiment Environment Setup

The experiment setting for the project is based on the use of Google Colaboratory (Colab) as the principal platform for development. This decision was taken because of its accessibility, collaborative features, and the availability of free GPU resources, all of which are crucial for the implementation of ML models and the execution of data analysis. In order to handle data preparation and model training, the system requirements include a reliable internet connection, a contemporary web browser that can access Google Colaboratory, and an enough amount of memory (RAM). The Python 3.12.0 version used for the project to guarantee compatibility with the most recent libraries and implementations.

Key tools that are integrated into the environment include Jupyter Notebooks, which are used for interactive development within Google Colab and a variety of Python libraries, including Pandas and NumPy for data manipulation; Matplotlib and Seaborn for data visualization; Scikit-learn for machine learning algorithms. The project makes use of NLTK and Spacy for text preprocessing, Wordcloud for visualizing word frequency patterns. All these tools are used within the framework of natural language processing (NLP)-based text mining. An additional factor to consider is the use of Google Drive for the smooth integration of data storage and the utilization of the free GPU resources that are available via Google Colab for the purpose of accelerating model training. For the purpose of ensuring readability and maintainability, the code structure places an emphasis on logical modules. Additionally, in-line comments and markdown cells in Jupyter Notebooks are used to provide comprehensive documentation of both the code and the methodology.

5.2. Data Analysis

In this phase of the project that dealt with the analysis of data, a comprehensive approach was chosen. This technique included the independent examination of three unique datasets which are described in above chapters which represented a different kind of fraud: cyberbullying, misinformation, and genuine reviews. The treatment of each dataset was carried out individually, which enabled a domain-specific comprehension of the fraud patterns that are present within the content of social media. Finally, an analysis was performed on the combined dataset to determine the link between the several predominant forms of fraud.

5.2.1. Analysis of Cyberbullying Datasets:

Dataset Introduction: As described above in chapter four this dataset is exported from kaggel database. In this dataset, there are nearly forty-seven thousand (~47,000) tweets that have been classified according to the different categories of cyberbullying. They have been labelled in five different type of bullying like religion, ethnicity, age, gender, and other forms of cyberbullying. It also includes non-cyberbullying tweet data.

Dataset Visualization:

The use of data visualization is very necessary when it comes to identifying instances of cyberbullying or any other fraud that occur on social media platforms. It does this by using visual components like as charts and graphs to find patterns, trends, and insights that are present in tweets that include cyberbullying. These graphics improve both exploratory and explanatory data analysis because of their ability to indicate frequency using bar charts and to showcase prominent phrases using word clouds. This condensed method aids decision-makers in efficiently combating social media fraud by transforming raw data into insights that can be put into action. In this paper, initially we have applied some of the tools for visualization to get closer to the datasets.

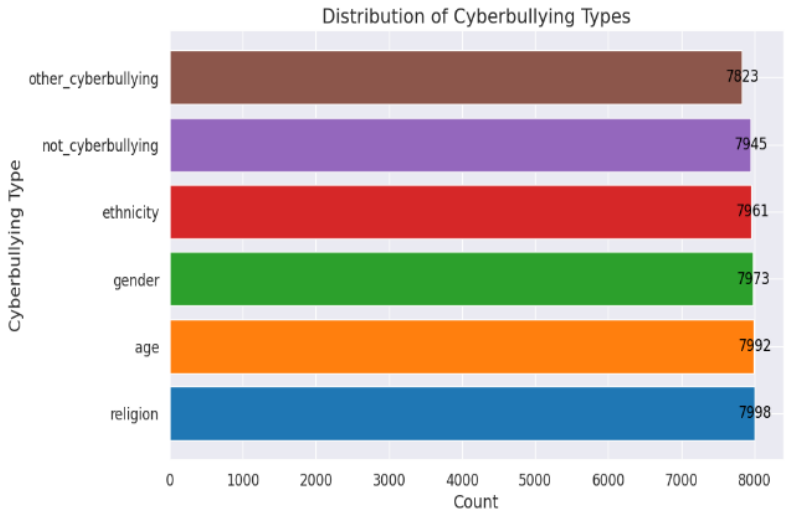


Figure 6: Distribution of different type of Cyberbullying in the Dataset.

According to the figure 6, it shows that the data has been adjusted such that there are about or nearly equal instances of data distribution in each category.

Word Clouds:

The size of the font and the color of the font are used to indicate the value of each individual word that is contained within a word cloud. An example of a visual representation of text data is called

a Word Cloud. Following that, we utilized the text pre-processing techniques that were outlined in chapter four. These techniques included converting all letters to lowercase, removing non-alphabetic characters, HTML tags, and URLs, removing white space, formatting contractions, tokenizing, stemming, and lemmatization. Finally, word clouds based on the dataset of cyberbullied tweets that had been cleared as followed based on type of bullying is generated .

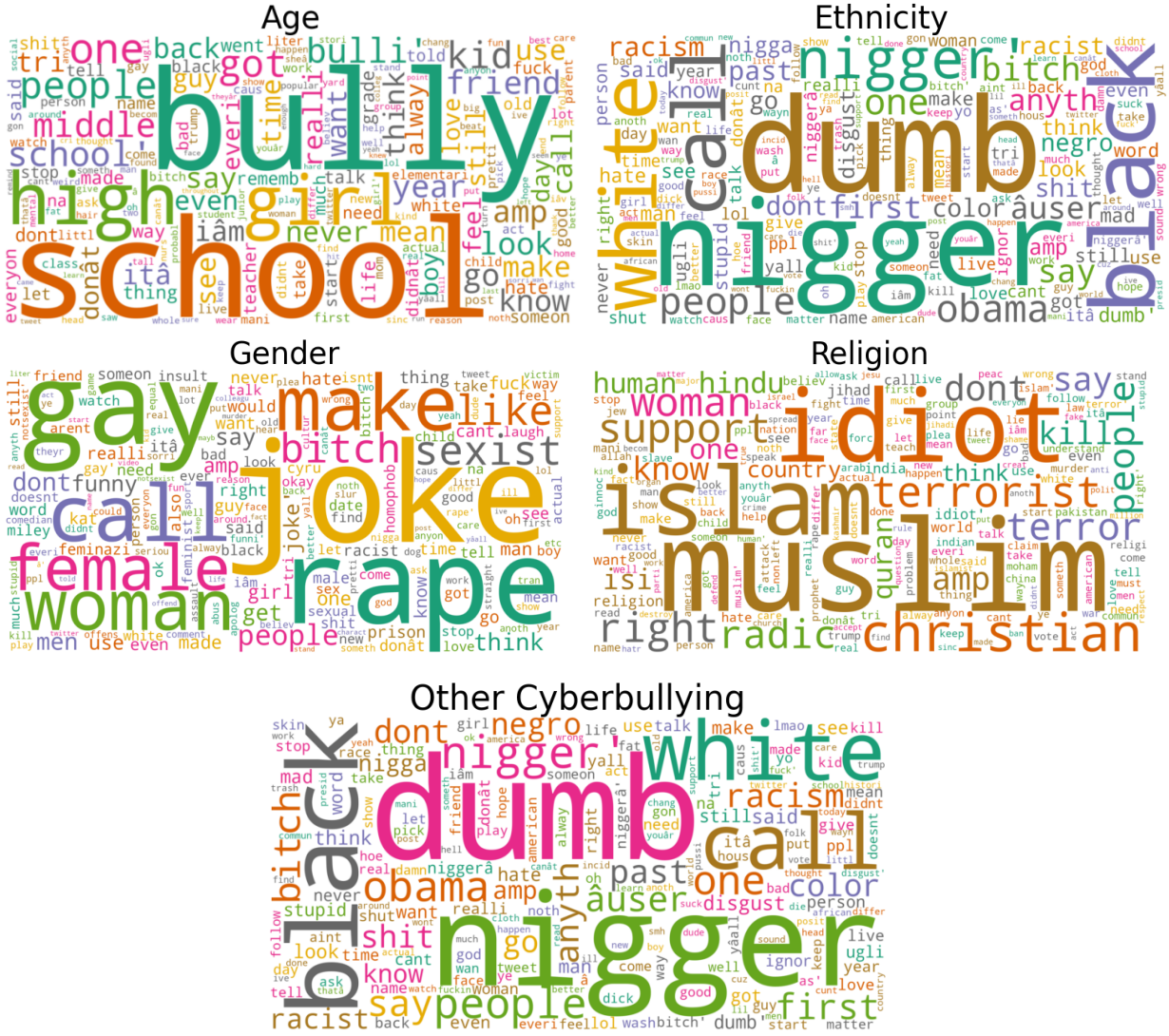


Figure 7: Word Clouds for different types of Cyberbullying Tweets.

The terms like "school," "bully," "high," "girl," in age based cyberbullying, “nigger”, ”dumb”, ”white”, ”black” in ethnicity based cyberbullying, “muslim”, “islam”, “idiot”, “christian” in religion based cyberbullying, “rape”, “joke”, “gay”, “woman” in gender base cyberbullying and finally, “bully”, “dont”, “make”, “think” and other similar phrases are considered to be the most significant in evaluations, as seen in figure 7. It can be flagged that tweets considering those words may be significant for fraud detection in social media.

Most used words and phrases:

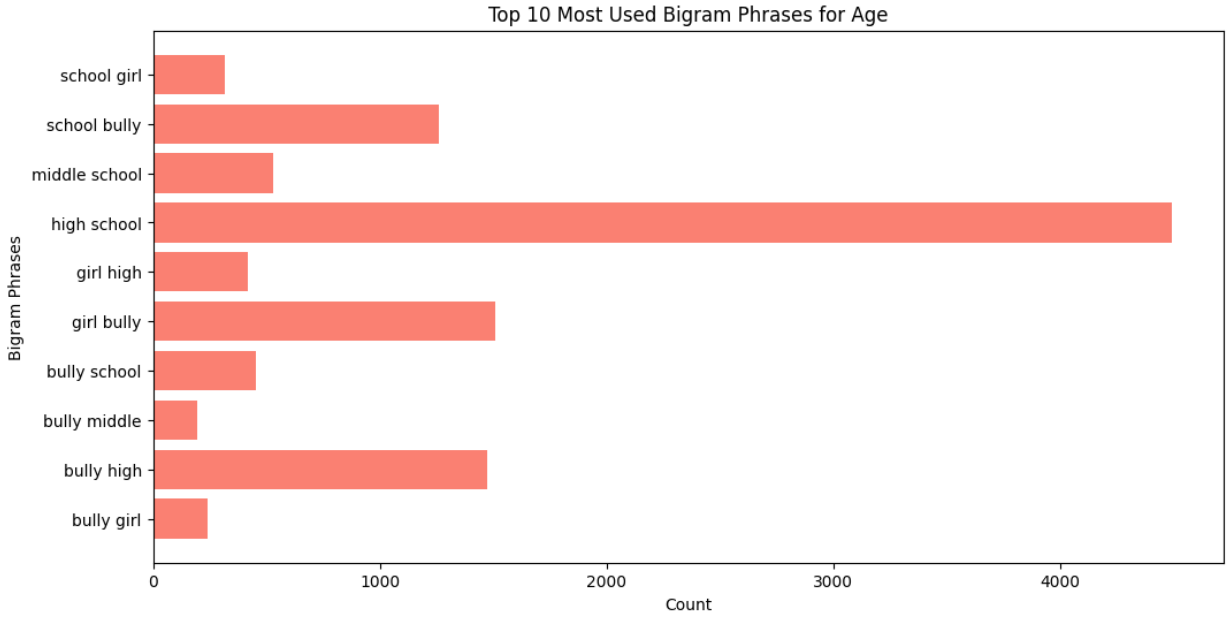


Figure 8: Top ten most used Bigram Phrases for Cyberbullying related to Age.

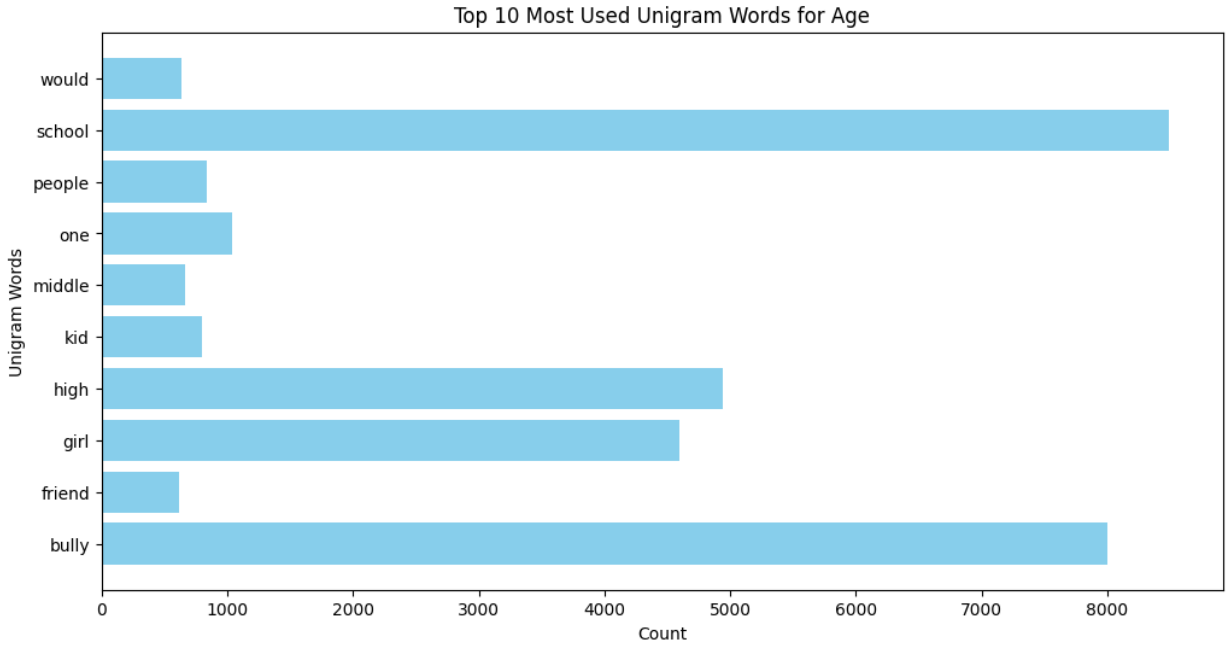


Figure 9: Top ten most used Unigram Words for Cyberbullying related to Age.

Figure 8 and figure 9 are examples of the frequency table, which is a bar graph, that illustrates the terms and phrases that have been used in tweets that are connected to cyberbullying in the age dominant demographic. The top ten most often used bigram phrases are shown in figure 8, and the top ten unigram terms are shown in figure 9. This example provides a basic concept of all the different types of cyberbullying that are included in datasets.

5.2.2. Analysis of Fake Reviews Datasets:

Dataset Introduction:

In a comparable manner, this dataset is about fake reviews, and it includes 20,000 fake reviews in addition to 20,000 real reviews of products. Detecting fraudulent reviews on online platforms was the focus of study conducted by Salminen et al. (2022), who developed and used this dataset for their investigation. It is made up of CG, which stands for computer-generated false reviews, and OR, which stands for original reviews, which are allegedly written by humans and are genuine as shown as example in Table 3.

Dataset Visualization:

The visualization of the dataset was carried out in the same manner as the example given above, namely in the cyberbullying dataset. In this case, an initial analysis of the dataset comprised of a pie chart that displayed a piece of each rating was performed.

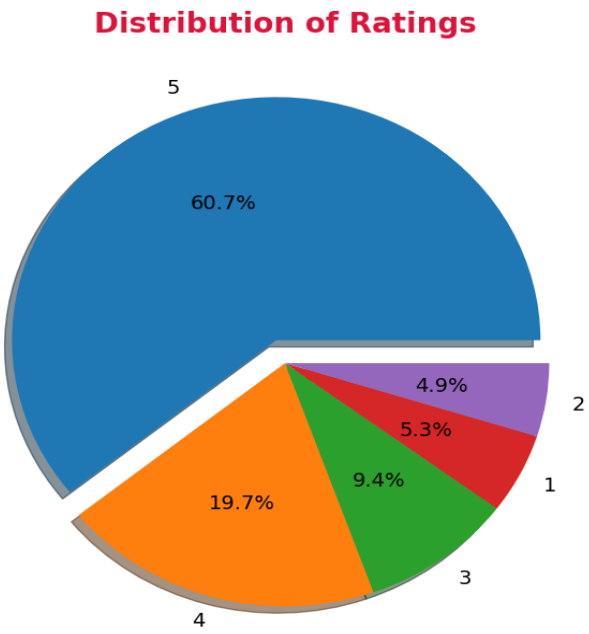


Figure 10: Distribution of Rating in the Fake Review dataset.

Word Cloud:

The word cloud of fake reviews is categorized into five distinct ratings. According to the data, the majority of false or computer-generated reviews exhibit a similar pattern. The most often used

terms for generating reviews are 'great', 'good', 'love', 'one', 'book', 'use', and others, as seen in figure 11.



Figure 11: Word Cloud for Fake Review with different Rating

Most used words and phrases:

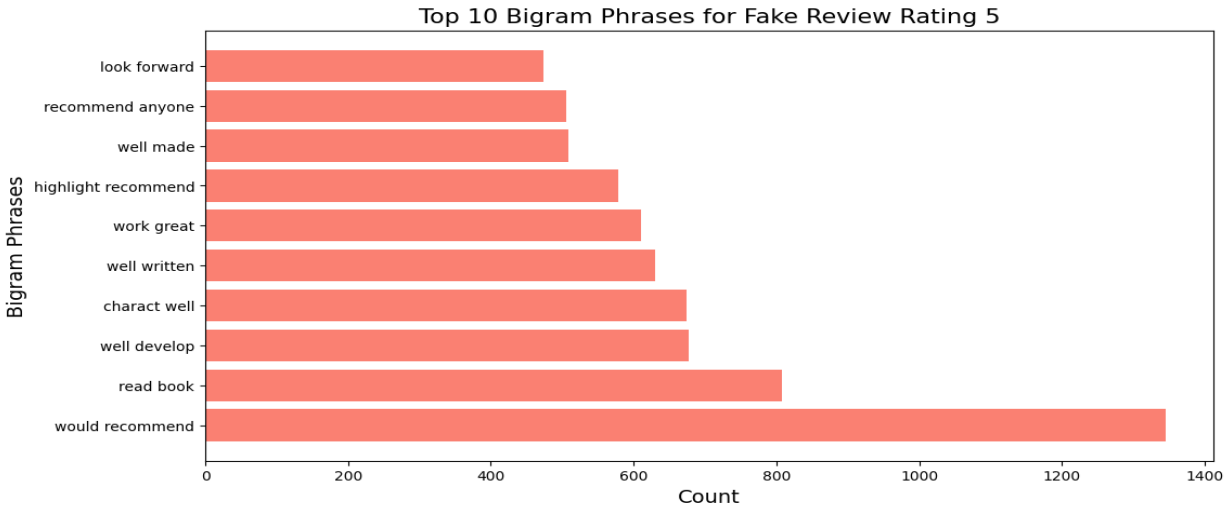


Figure 12: Top ten Bigram Phrases for Fake Review with Rating Five

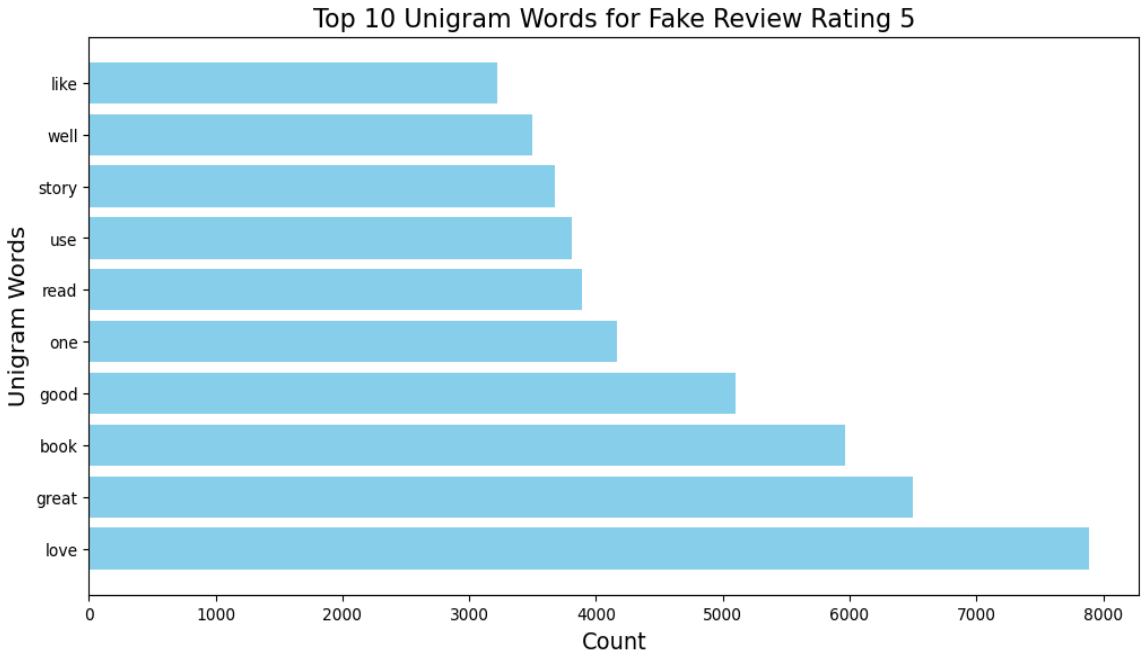


Figure 13: Top ten Unigram Words for Fake Review Rating Five

The frequency table, shown as a bar graph in Figures 12 and 13, shows the different words with and without phrases that have been used in twitter dataset related to fake reviews in the rating five category. Figure 12 displays the top ten frequently used bigram phrases, which include "look forward", "recommend anybody" "well made", "highlight recommend", "work great", "well written", "charact well", "well develop", "read book" and "would recommend". On the other hand, figure 13 presents the top ten unigram terms, which consist of "like," "well," "story," "use," "read," "one," "good," "book," "great," and "love." This example presents a fundamental understanding of the many sorts of language used in deceptive or computer-generated evaluations to signify deceit in social media.

5.2.3. Analysis of Misinformation Datasets:

Dataset Introduction:

This dataset relates to the collection of tweets about disasters. The dataset aims to simplify the process of identifying relevant and non-relevant data about disasters from social media. The dataset comprises of tweets that have been manually categorized into two distinct groups: relevant and irrelevant. Out of a total of 10,877 distinct tweets, 57% are deemed irrelevant while the remaining 42% are considered relevant. The example of dataset is shown in Table 4.

Dataset Visualization:

Initially, we examined the distribution of classes in the dataset by utilizing a plot diagram depicted in Figure 1. This diagram provides a visual representation of the distribution of tweets that are either related to disasters or unrelated to disasters. It reveals that 57% of the tweets are classified as unreal disasters, while the remaining 43% are associated with real disasters. This suggests that the classes are evenly distributed, which is advantageous for both training and assessing the NLP model.

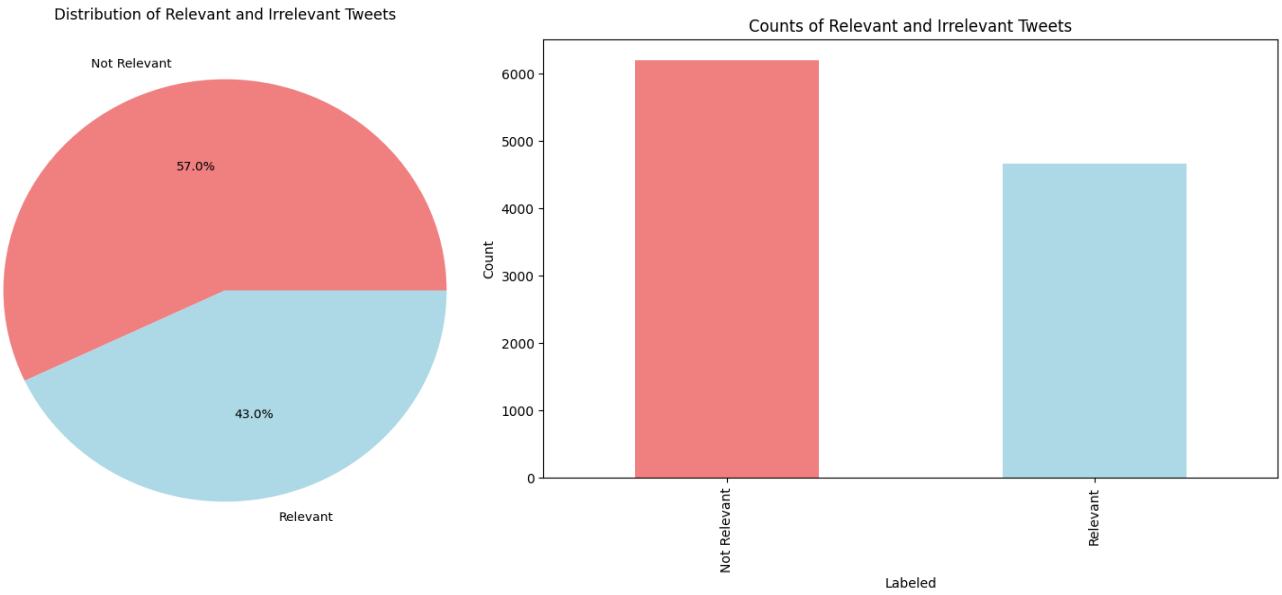


Figure 14: Distribution and Counts for Relevant and Irrelevant Tweets in Dataset

Word Cloud:

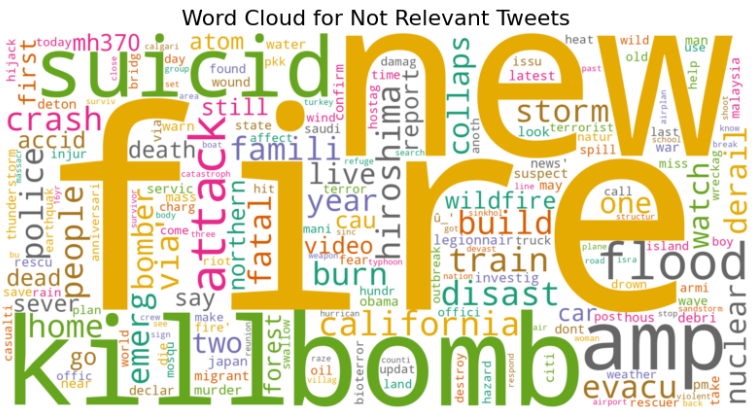


Figure 15: Word Cloud for Not Relevant Tweets.

In Figure 15, the real word cloud of the irrelevant tweets is shown in relation to misleading tweets. This technique validates the efficacy of using key terms such as "fire," "new," "kill," "bomb,"

"suicide," and other similar keywords to forecast the presence of irrelevant data in the dissemination of misleading information during instances of disaster.

Most used words and phrases:

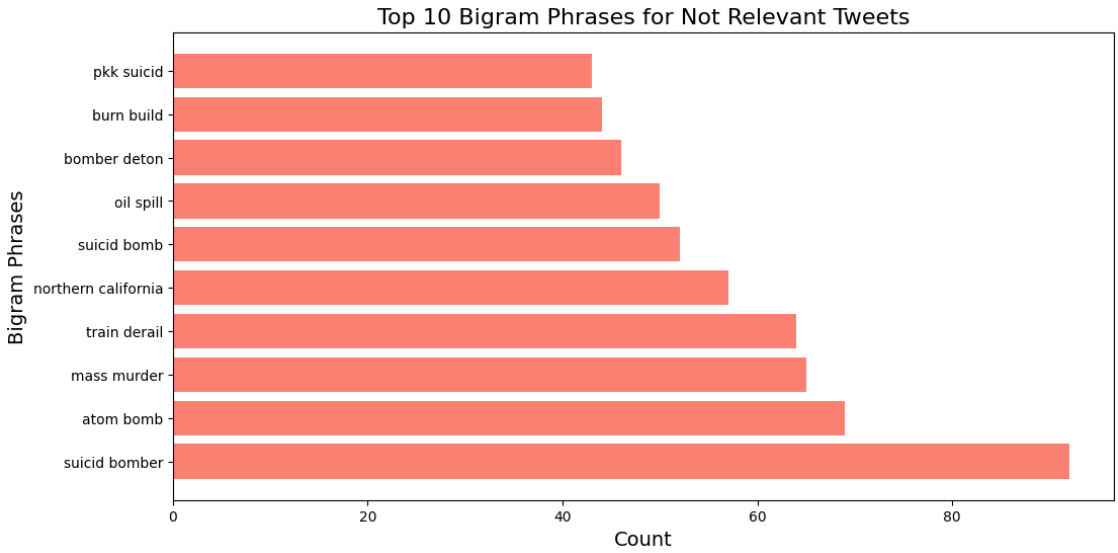


Figure 16: Top ten Bigram phrases for not Relevant Tweets in misleading information about Disaster

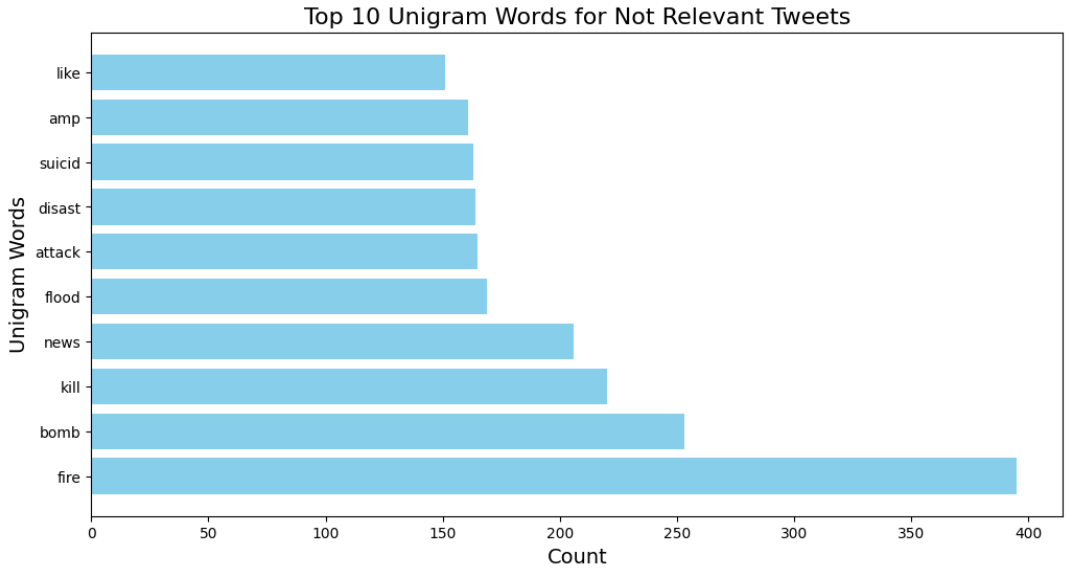


Figure 17: Top ten Unigram phrases for not Relevant Tweets in misleading information about Disaster

5.3. Feature Extraction

"Feature extraction" refers to the process of converting raw data into numerical characteristics that can be processed while preserving the information that was included in the initial data set.

This process is carried out to maintain the integrity of the data. As a result of the superior results it generates, it is more efficient than applying machine learning directly to the raw data..

Feature extraction using Unigram and Bigram with TFIDF:

Feature extraction using unigram TF-IDF along with bigram TF-IDF requires gathering several levels of textual information to represent texts in a numerical representation for the purpose of machine learning models.

When deciding between unigram TF-IDF and bigram TF-IDF, the unique qualities of the text data involved as well as the job at hand should be taken into consideration. Unigrams are useful for catching generic word occurrences, although bigrams may give more information and record specific language patterns. Unigrams are often used in linguistic analysis. In fact, it may be possible to employ a mix of unigrams and bigrams, also known as N-grams in general, to achieve a balance between the granularity of information and the contextual information.

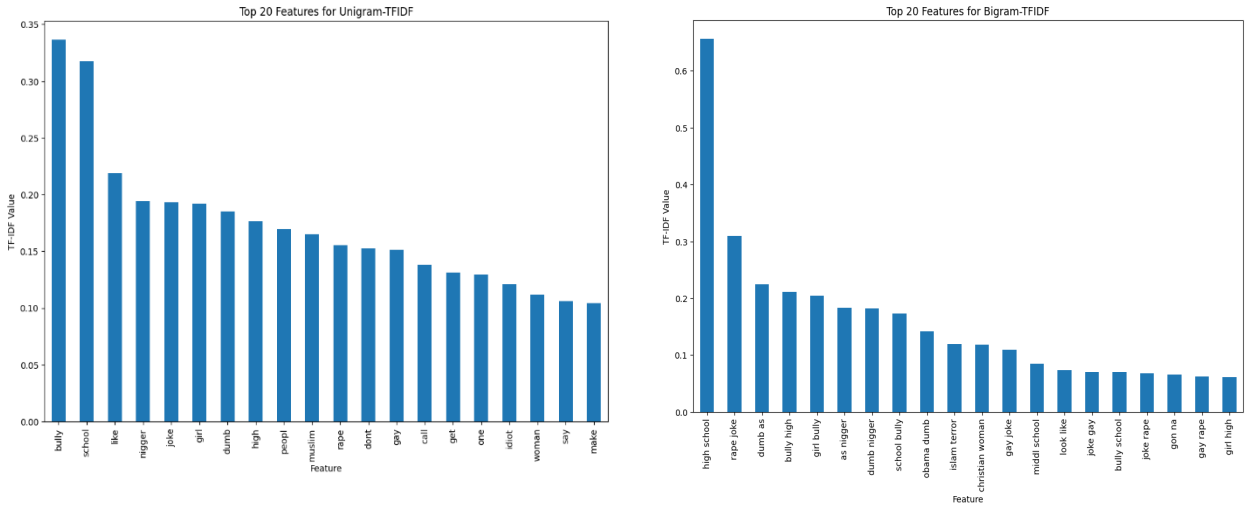


Figure 18: Top 20 feature representation for Bigram TF-IDF and Unigram TF-IDF in Cyberbullying Dataset.

According to figure 18, the top twenty features for Unigram TF-IDF analysis highlight the importance of certain terms in the text, such as "bully" and "school," which have high TF-IDF values of 0.336333 and 0.317588, respectively. The findings suggest that certain terms have a high frequency inside the text and are also infrequent across other documents, highlighting their uniqueness. Additional influential words such as "like," "nigger," and "joke" enhance the comprehensive comprehension of essential characteristics. The TF-IDF values provide insights into the distinctiveness and significance of these specific terms, with larger values indicating a greater influence within the dataset.

On the other hand, the Top 20 Features for Bigram TF-IDF analysis focus on pairs of neighboring words, as seen by the highest TF-IDF score of 0.656292 for the phrase "high school". This indicates that the bigram "high school" is not only often found inside the text, but also seldom found across numerous papers, making it a significant trait. Significant contextual meaning is conveyed by notable bigrams such as "rape joke," "dumb as," and "bully high." For instance, the phrase "obama dumb" may allude to certain viewpoints on Obama, whilst "islam terror" might highlight conversations pertaining to Islam and terrorism. The comparison highlights that Unigram TF-IDF emphasizes the influence of individual words, whereas Bigram TF-IDF considers the collective importance of word pairs, offering a thorough comprehension of unique characteristics and their contextual value in the dataset.

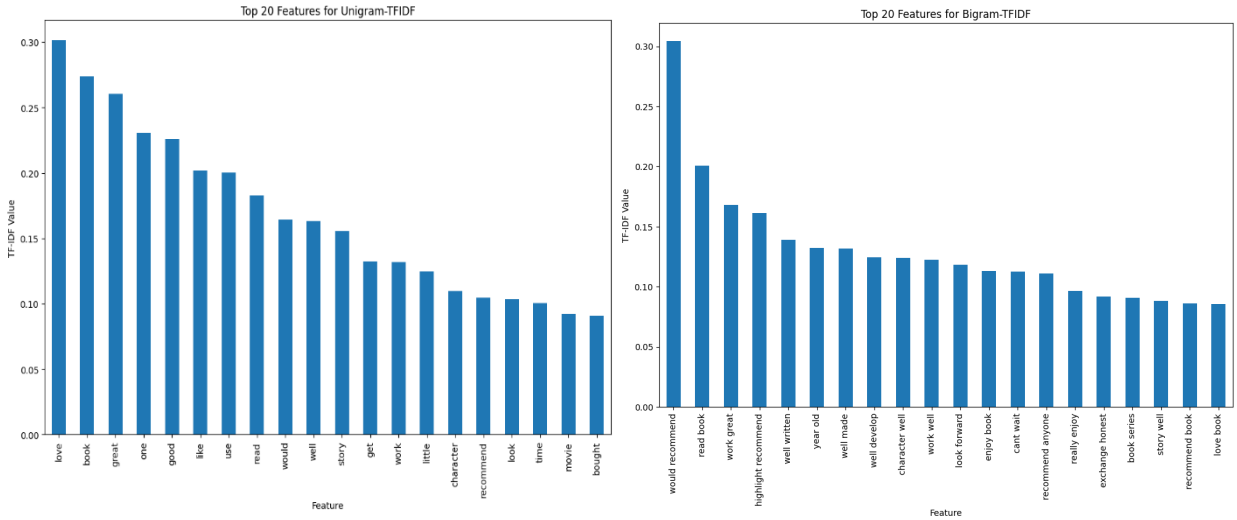


Figure 19: Top 20 feature representation for Bigram TF-IDF and Unigram TF-IDF in Fake Review Dataset.

Similarly, analyzing figure 19, in case of fake reviews dataset the leading 20 features for Bigram TF-IDF and Unigram TF-IDF examinations provide a comprehensive insight into textual features as cyberbullying which can be explained from figure 19 as well. In the context of bigrams, the high TF-IDF score of 0.304278 for the phrase "would recommend" indicates its frequent occurrence in the text and its infrequent appearance in other papers, highlighting its significance. Additional influential bigrams such as "read book," "work great," and "well written" provide subtle contextual importance to the text. For example, the phrase "highlight recommend" may indicate suggestions, while "well develop" emphasizes the feature of growth and progress. In addition, the expression of positive emotions is expressed via bigrams such as "look forward," "enjoy book," and "can't wait," which indicate both the pleasure and anticipation connected with these terms. When considering individual words, the greatest TF-IDF value of 0.301710 for the word "love" indicates that it appears often in the text but is rare in other papers, making it distinct.

Likewise, the words "book," "great," and "good" have significant TF-IDF values, highlighting their frequency and significance. The phrase "one" is significant, and terms such as "like," "use," and "read" add to a full comprehension of fundamental characteristics. Moreover, phrases like as "well," "story," and "character" provide valuable insights on certain themes or attributes that are evident in the text. The TF-IDF values quantify the distinctiveness and significance of certain words and word pairs in the dataset, hence offering useful insights on frequent themes or expressions.

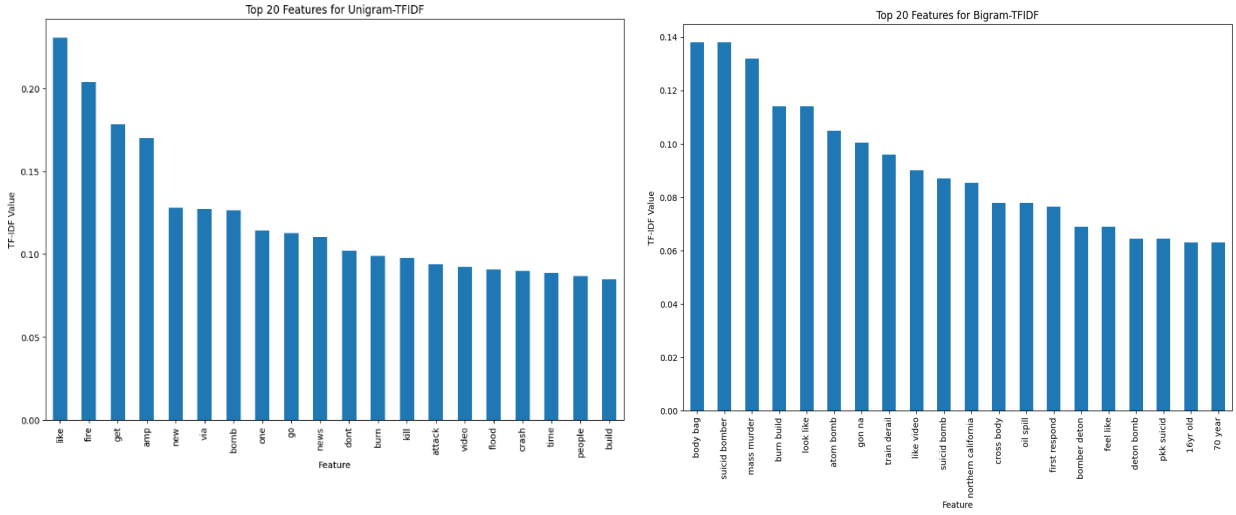


Figure 20: Top 20 Features for Unigram TF-IDF and Bigram TF-IDF in Misinformation Dataset

Finally, figure 20 shows the misinformation dataset on disaster the analysis of the top 20 features, including both bigram TF-IDF and unigram TF-IDF as shown in figure 20 above, reveals a complex web of meaning made up of adjacent word pairs and individual words in the analyzed text. In the Bigram TF-IDF domain, the pair “body bag” and “suicide bomber” shines with a common his TF-IDF value of 0.137937, indicating their uniqueness and importance. These phrases, along with notable bigrams such as "mass murder" and "incendiary construction," contribute to the story contextually. The inclusion of bigrams such as "atomic bomb," "train derailment," and "oil spill" indicates a discussion of disasters and emergencies. “Northern California” introduces geographic context, and “crossbody” suggests a specific action or scenario. Additionally, bigrams such as "mood" and "16 years old" provide insight into discussions related to emotions and age.

Unigram TF-IDF focuses on individual words. Like has attracted attention with his TF-IDF value of 0.230610, indicating its prevalence and uniqueness. Words such as "fire," "get," and "amp" greatly contribute to understanding key functions. The term "new" in 0.127855 suggests discussion about current events, while "bomb," "attack," and "kill" deal with topics related to disasters and conflicts. Video and News highlight multimedia content and discussions about

current topics. “Floods” and “Crashes” provide insight into specific disaster-related topics. The TF-IDF value serves as a quantitative measure, highlighting the uniqueness and relevance of these words. Higher values indicate greater influence in the dataset. This dual analysis provides a better understanding of key themes, contextual nuances, and meanings of specific words, providing a more comprehensive understanding of the content of the dataset.

5.4. Model Fitting and Evaluation

In this step of analysis, the dataset is firstly split into train and test sets in the early stages of the model fitting with above mention features using the Naive Bayes method. Typically, this approach uses an 80%–20% split. This method uses unigram, bigram and n-gram approach separately for the model fitting purpose for analysis of their performance. Dividing the data is essential for evaluating the model's performance on data that it has not been trained on. The features that have been processed are used as the basis for training the Naive Bayes model. Throughout the training phase, the algorithm acquires knowledge of the fundamental patterns in the training set, linking the individual word characteristics with the relevant labels that indicate whether a text is related to different type of fraud using all three datasets to predict if the data is frauded or not. This stage is crucial for the model to provide accurate predictions on novel, unknown texts during the assessment phase. The accuracy of the succeeding model is highly dependent on the efficacy of the data splitting and preprocessing steps, since they directly influence the quality of the features used for training the Naive Bayes classifier.

5.4.1. Model Evaluation with Unigram Features

```
Accuracy: 0.7279589055456547
Classification Report:
                    precision    recall  f1-score   support

               age       0.69      0.98      0.81      1603
         ethnicity       0.82      0.88      0.85      1603
            gender       0.75      0.82      0.78      1531
 not_cyberbullying       0.69      0.32      0.43      1624
other_cyberbullying       0.61      0.42      0.50      1612
          religion       0.75      0.97      0.85      1566

          accuracy                           0.73      9539
         macro avg       0.72      0.73      0.70      9539
      weighted avg       0.72      0.73      0.70      9539
```

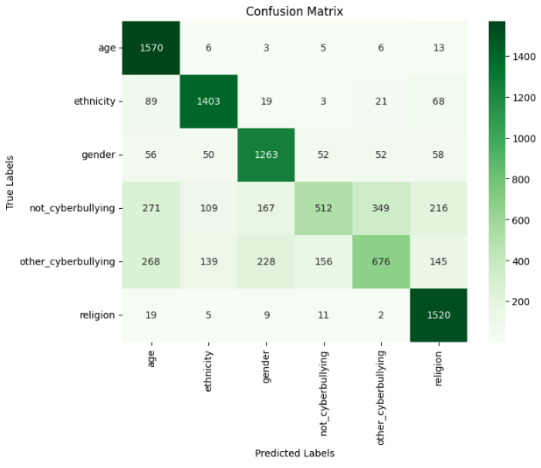


Figure 21: Confusion Matrix, Accuracy and Classification Report on Unigram Features in Cyberbullying Dataset.

An overall accuracy of around 72.8% is shown by the assessment metrics for the Naive Bayes classifier that was trained on unigram TF-IDF features throughout the training process. Additionally, the categorization report provided in Figure 21 gives an in-depth analysis of the performance of the model across a variety of categories. When it comes to predicting 'age' and 'religion' classes, the model performs very well, reaching accuracy and recall values that are more than 0.75. On the other hand, it may have difficulty distinguishing between cases of "not cyberbullying" and "other cyberbullying," which is a situation in which the trade-off between accuracy and recall is obvious, which ultimately leads to lower F1-scores. There are class imbalances that are taken into consideration by the weighted average, which highlights the overall efficacy of the model across the many categories that are included in the dataset. These measurements provide a full picture of the model's strengths and shortcomings, hence leading prospective adjustments that might lead to improved performance in certain courses.

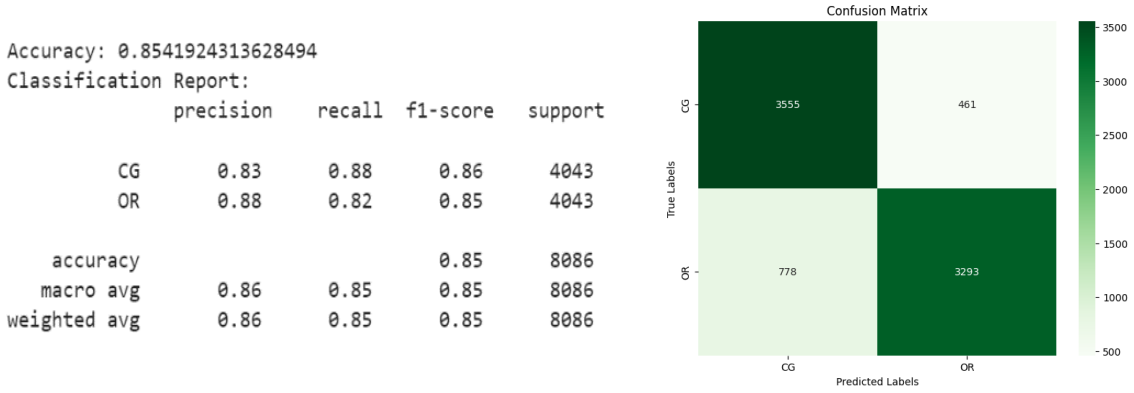


Figure 22: Confusion Matrix, Accuracy and Classification Report for Model with Unigram Features in Fake Reviews Dataset.

The findings shows in Figure 22, that the model is successful in recognizing fake reviews, as it achieved an excellent accuracy of 85.42% across the board. In other words, the model properly identified more than 85 percent of all the reviews that were included in our dataset. When we take a closer look at precision, which is a measurement of how accurate our model is when it predicts a certain kind of review, we find that the model is accurate around 83% of the time when it comes to computer-generated (CG) reviews.

When it comes to original reviews (OR), the precision rises to 88%, which indicates that the reviews have a high degree of correctness. For CG, the recall is 88%, while for OR, it is 82%. The recall is a measurement of the model's capacity to recognize all occurrences of a certain kind of review. It seems from these numbers that the model is especially successful when it comes to identifying reviews that were created by a machine. CG has an F1-score of 86%, whereas OR has an F1-score of 85%. This number indicates a compromise between accuracy and recall. This

exemplifies how both sorts of evaluations are characterized by a complementary combination of accuracy and memory. There are a total of 4043 occurrences of both CG and OR reviews included within the dataset. The model's solid success in discriminating between computer-generated and authentic evaluations is reinforced by the fact that the macro and weighted averages, which combine to create an overall assessment, show an accuracy of 85%.

```
Accuracy: 0.8000919117647058
Classification Report:
              precision    recall  f1-score   support

Not Relevant       0.77      0.91      0.84      1214
    Relevant       0.85      0.66      0.75       962

    accuracy                           0.80      2176
   macro avg       0.81      0.79      0.79      2176
weighted avg       0.81      0.80      0.80      2176
```

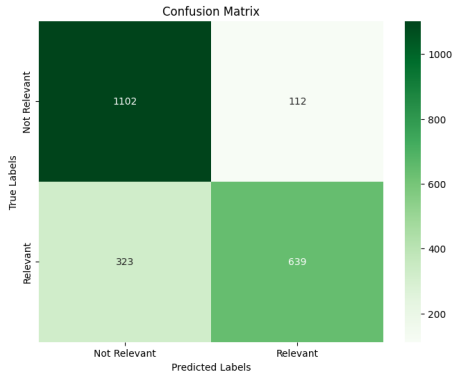


Figure 23: Confusion Matrix, Accuracy and Classification Report for Model with Unigram Features in Misinformation Dataset

Also, the result presented in Figure is a preview of the performance metrics of a model that was trained on a dataset that included information on tweets regarding disasters, with a special emphasis on the identification of tweets that matched the criteria for relevance. It is estimated that the model has an accuracy of roughly 80.01%, which indicates that it accurately identified the category of nearly 80% of the tweets included in the dataset.

When the classification report is split down into its component parts, precision is a measurement that determines how accurate positive predictions are. In this instance, it shows that the model is accurate around 85 percent of the time when it labels a tweet as relevant. Recall, which is a measurement of the model's capacity to catch all relevant occurrences, is about 66%, which indicates that the model detected 66% of all tweets that meet the criteria for relevance. 0.75 is the value of the F1-score, which indicates a compromise between accuracy and recall.

There are 1214 instances of tweets that have been categorized as "Not Relevant," and there are 962 occurrences that have been labeled as "Relevant." This is based on the support values. When it comes to the overall performance of the model across classes, the macro average and weighted average give valuable comprehensive information.

In conclusion, while the model exhibits a high level of accuracy and precision, there is space for development in terms of recall, which indicates that it might effectively collect tweets that are

relevant to the topic at hand. There is a possibility that the performance of the model might be improved by doing further optimization or considering the goals of the study.

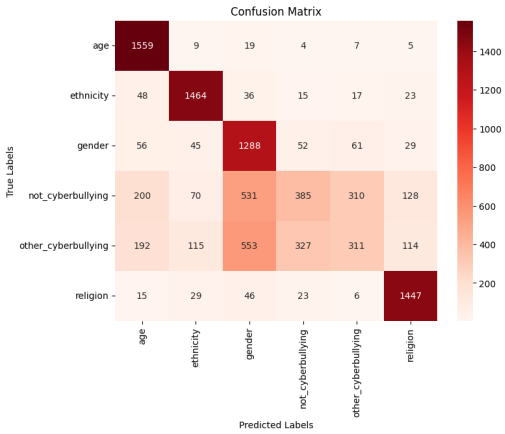
5.4.2. Model Evaluation with Bigram Features

The Bigram Featured model fitting on the cyberbullying dataset yielded an accuracy of roughly 67.66% according to the experiment as shown in figure 24. The accuracy for classifying various categories varies, with a high precision of 75% for the "age" class, 85% for the "ethnicity" class, and 83% for the "religion" class. This indicates that the model accurately predicts tweets belonging to these categories a considerable fraction of the time. The recall rate for these courses is often high. Nevertheless, the accuracy and recall for the classes "not_cyberbullying" and "other_cyberbullying" are comparatively lower, suggesting difficulties in accurately detecting and recording occurrences of these classes. The macro-average F1-score is around 0.64, indicating a modest equilibrium between accuracy and recall across several classes. Although the model demonstrates satisfactory performance in some areas, there is still potential for improvement, particularly in accurately differentiating between cases of cyberbullying and non-

```
Accuracy: 0.6765908376140056
Classification Report:
                     precision    recall  f1-score   support

                age       0.75      0.97      0.85      1603
          ethnicity       0.85      0.91      0.88      1603
             gender       0.52      0.84      0.64      1531
 not_cyberbullying       0.48      0.24      0.32      1624
other_cyberbullying       0.44      0.19      0.27      1612
           religion       0.83      0.92      0.87      1566

           accuracy                           0.68      9539
          macro avg       0.64      0.68      0.64      9539
       weighted avg       0.64      0.68      0.64      9539
```



cyberbullying.

Figure 24: Confusion Matrix, Accuracy and Classification Report for Model with Bigram Features in Cyberbullying Dataset.

Below figure 25 show the Bigram Featured model fitting on tweets with fake reviews yielding an accuracy of roughly 72.64%. The accuracy in detecting "CG" (computer-generated or false) tweets is 65%, meaning that the algorithm correctly identifies a tweet as "CG" 65% of the time. The recall rate for "CG" tweets is 97%, indicating that the model successfully identifies the bulk of genuine "CG" tweets. Nonetheless, the accuracy in recognizing "OR" (original) tweets is greater at 95%, indicating that when the algorithm classifies a tweet as "OR" it is accurate 95% of the time. The recall rate for "OR" tweets is 48%, suggesting that the algorithm may fail to identify some genuine "OR" tweets. In general, the model exhibits satisfactory performance in

differentiating between computer-generated and original tweets, however there is a compromise between accuracy and recall for each category.

```
Accuracy: 0.7263509335971312
Classification Report:
              precision    recall  f1-score   support

          CG       0.65      0.97      0.78      4016
          OR       0.95      0.48      0.64      4071

    accuracy                           0.73      8087
   macro avg       0.80      0.73      0.71      8087
weighted avg       0.80      0.73      0.71      8087
```

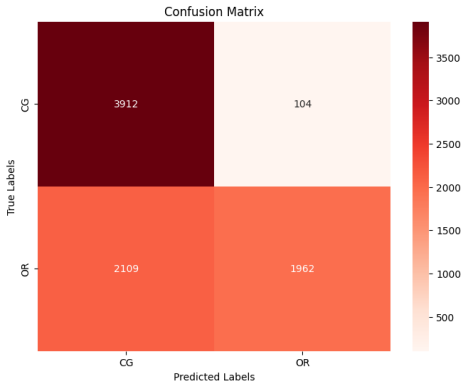


Figure 25: Confusion Matrix, Accuracy and Classification Report for Model with Bigram Features in Fake Reviews Dataset

Similarly, the accuracy of the Bigram Featured model fitting on misinformation in tweets about disasters is about 75.97%, according to the results of the fitting as shown in figure 26. The precision for recognizing tweets that are "Not Relevant" is 73%, which means that when the model makes a prediction that a tweet is "Not Relevant," it is accurate 73% of the time because of the precision. The recall rate for tweets that are categorized as "Not Relevant" is 91%, which indicates that the model understands a significant amount of the tweets that are classified as "Not Relevant." On the other hand, the precision for detecting tweets that are "Relevant" is 83%, which means that when the model predicts a tweet to be "Relevant," it is accurate 83% of the time. The recall rate for tweets that are considered "Relevant" is 57%, which indicates that the model may overlook some tweets that are really "Relevant." When it comes to discriminating between tweets that are relevant and tweets that are not relevant, the model does rather well overall; nevertheless, there is potential for improvement in terms of recall for tweets that are relevant.

```
Accuracy: 0.7596507352941176
Classification Report:
              precision    recall  f1-score   support

Not Relevant       0.73      0.91      0.81      1214
    Relevant       0.83      0.57      0.68       962

    accuracy                           0.76      2176
   macro avg       0.78      0.74      0.74      2176
weighted avg       0.77      0.76      0.75      2176
```

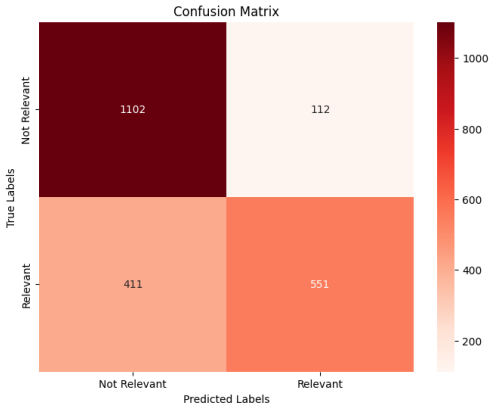
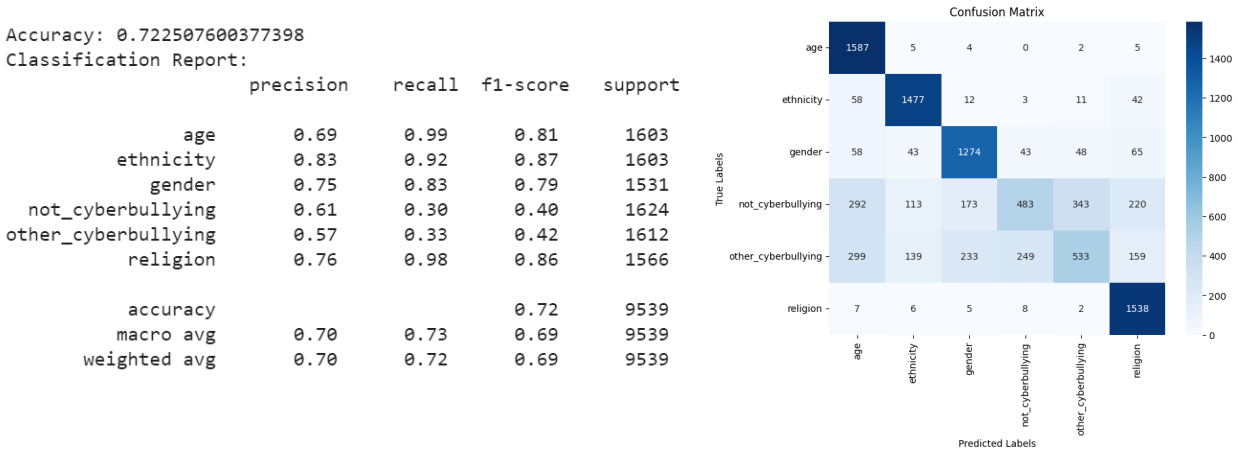


Figure 26: Confusion Matrix, Accuracy and Classification Report on Model with Bigram Features in Misinformation Dataset.

5.4.3. Model Evaluation with Combined Uni and Bigram Features

The model, which included a combination of unigram and bigram features, obtained an overall accuracy of 72.25% when trained on a cyberbullying dataset. The precision, which assesses the correctness of positive predictions, varied from 57% to 83% across various categories, suggesting that the model typically makes accurate predictions for certain categories. Nevertheless, the recall, which measures the model's capacity to correctly identify all occurrences of a certain class, exhibited variability across different classes. The highest recall rate was seen for the "age" class, reaching an impressive 99%. Conversely, the "not_cyberbullying" class had the lowest recall rate, standing at a mere 30%. The F1-scores, which consider both accuracy and recall, varied between 40% and 87%. Although the model exhibits satisfactory performance, particularly in classes such as "age," "ethnicity," and "religion," there is scope for improvement, notably in reliably detecting cases of "not_cyberbullying." Additional optimization or experimentation with other models may improve overall performance.



Accuracy: 0.722507600377398
Classification Report:

	precision	recall	f1-score	support
age	0.69	0.99	0.81	1603
ethnicity	0.83	0.92	0.87	1603
gender	0.75	0.83	0.79	1531
not_cyberbullying	0.61	0.30	0.40	1624
other_cyberbullying	0.57	0.33	0.42	1612
religion	0.76	0.98	0.86	1566
accuracy			0.72	9539
macro avg	0.70	0.73	0.69	9539
weighted avg	0.70	0.72	0.69	9539

Figure 27: Confusion Matrix, Accuracy and Classification Report for Combined Unigram and Bigram Features for Cyberbullying Dataset.

The model used a combination of unigram and bigram features on a dataset of tweets that included fake reviews, resulting in an outstanding accuracy rate of 86.34%. The precision of positive predictions, which measure the accuracy of distinguishing computer-generated (CG) or original (OR) evaluations, ranged from 82% to 92%, demonstrating that the model is accurate in this task. The recall, which measures the model's capacity to accurately detect all occurrences of a class, varied from 79% to 93%, demonstrating the model's efficacy in properly recognizing instances of both CG and OR reviews. The F1-scores, which represent a balance between accuracy and recall, showed a noteworthy level of performance, ranging from 85% to 87%. In general, the model exhibits strong performance in differentiating between CG and OR evaluations, offering dependable predictions with a well-balanced compromise between accuracy and recall.

```
Accuracy: 0.8633609496723136
Classification Report:
              precision    recall  f1-score   support

          CG       0.82      0.93      0.87      4016
          OR       0.92      0.79      0.85      4071

    accuracy                           0.86      8087
   macro avg       0.87      0.86      0.86      8087
weighted avg       0.87      0.86      0.86      8087
```

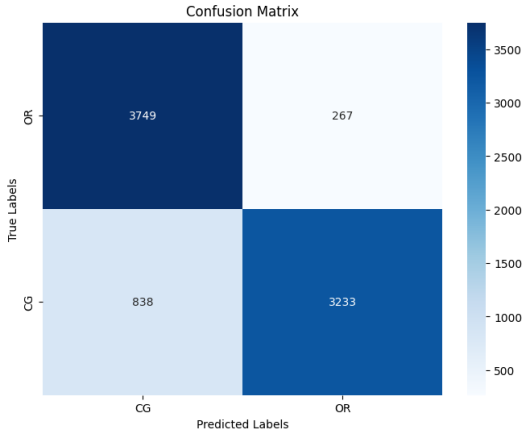


Figure 28: Confusion Matrix, Accuracy and Classification Report for Combined Unigram and Bigram Features for Fake Review Dataset

The model achieved an accuracy of 80.79% in categorizing tweets related to disaster information, exhibiting a well-balanced performance in terms of precision, recall, and F1-score. Regarding precision, which evaluates the correctness of positive predictions, the model demonstrated good outcomes. The accuracy for tweets classified as "Not Relevant" was 78%, suggesting that the algorithm accurately identified a tweet as not relevant 78% of the time. The accuracy rate for "Relevant" tweets was 86%, indicating a robust capability to precisely detect disaster-related information that is pertinent. The recall, which measures the model's capacity to correctly identify all occurrences of a class, had a significantly high performance for both the "Not Relevant" and "Relevant" categories, with values of 91% and 68% respectively. This suggests that the model successfully detected a significant part of pertinent tweets while also keeping a strong ability to distinguish non-pertinent ones. The F1-score, which accounts for both accuracy and recall, exhibited an equitable compromise between the two, suggesting the model's proficiency in categorizing tweets connected to disasters.

```
Accuracy: 0.8079044117647058
Classification Report:
              precision    recall  f1-score   support

Not Relevant       0.78      0.91      0.84      1214
    Relevant       0.86      0.68      0.76       962

    accuracy                           0.81      2176
   macro avg       0.82      0.79      0.80      2176
weighted avg       0.82      0.81      0.80      2176
```

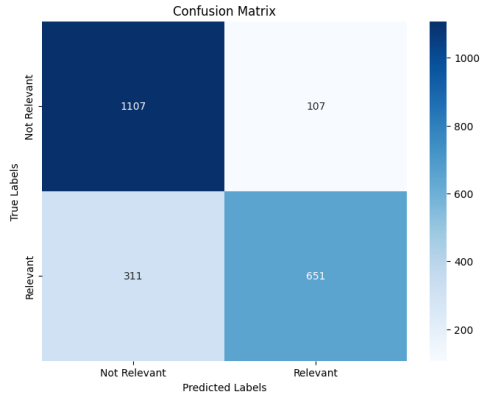


Figure 29: Confusion Matrix, Accuracy and Classification Report for Combined Unigram and Bigram Features for Misinformation Dataset

CHAPTER SIX: RESULTS AND DISCUSSION

6.1. Analysis of Results

According to the result summarized in Table 8, it shows the performance metrics of three distinct feature sets bigram, unigram, and combined unigram-bigram on a dataset focused on Cyberbullying. The unigram technique achieved a superior accuracy rate of 73%, demonstrating the model's proficiency in accurately predicting class labels for most occurrences. The results of this strategy showed higher levels of accuracy, recall, and F1-Score in comparison to the other feature sets. Specifically, the precision, recall, and F1-Score values were 71.8%, 81.4%, and 75.8%, respectively. The use of pairs of consecutive words as features, known as the Bigram technique, resulted in a lower accuracy rate of 68%.

However, it showed a considerably higher recall rate of 76.6%, indicating its effectiveness in identifying positive cases. The incorporation of both individual words and word pairs in the combined unigram-bigram technique yielded a balanced performance, achieving an accuracy of 72%. The precision, recall, and F1-Score values were moderate, with precision at 72%, recall at 79.8%, and F1-Score at 75%. The selection of the most suitable feature set is contingent upon the objectives of the classification assignment. The unigram technique demonstrates superior performance across all aspects, while the bigram approach places emphasis on recall. On the other hand, the Combined approach provides a middle ground between the two.

Furthermore, the selection of the most appropriate feature set should be influenced by factors such as the features of the dataset, the distribution of classes, and the available computing resources.

Table 8: Result of Performance of different Featured based Model in Cyberbullying Dataset.

Feature	Dataset	Accuracy	Average Precision	Average Recall	Average F1-Score
Bigram	Cyberbullying	68%	67.8%	76.6%	70.2%
Unigram	Cyberbullying	73%	71.8%	81.4%	75.8%
Combined	Cyberbullying	72%	72%	79.8%	75%

Similarly, table 9 provides an overview of the performance metrics for three unique feature sets: Bigram, Unigram, and Combined Unigram-Bigram, which were applied to a dataset that was primarily concerned with identifying false reviews. The Bigram method, which makes use of pairs of consecutive words as features, was able to reach an accuracy of 73% in this instance. It is noteworthy that the model had a high recall of 97%, which indicates that it is capable of accurately recognizing a considerable fraction of genuine positive events. On the other hand, the accuracy

was somewhat lower, coming in at 65%, which suggests that there was a larger proportion of false positives. The F1-Score, which is a measure that considers both accuracy and recall, came out to be 78%. A greater overall accuracy of 85% was shown using the Unigram method, which makes use of individual words as features. It achieved a performance that was well-balanced, with accuracy and recall coming in at 83% and 88%, respectively, which resulted in an exceptional F1-Score of 86%. The Combined Unigram-Bigram method, which combines individual words as well as word pairs, was successful in achieving an accuracy of 86%. A high F1-score of 87% was realized because of this strategy, which indicated a balanced trade-off between precision (82%), recall (93%), and overall accuracy. Particularly noteworthy is the fact that the Bigram technique was beaten by both the Unigram and Combined approaches across key parameters, with the Unigram approach demonstrating a solid performance on this fake review detection dataset. It was found that the Combined strategy, which made use of both individual words and word pairs, was successful in achieving a balance between accuracy and recall.

Table 9: Result of Performance of different Featured based Model in Fake Review Dataset.

Feature	Dataset	Accuracy	Precision	Recall	F1-Score
Bigram	Fake Review	*73%*	*65%*	*97%*	*78%*
Unigram	Fake Review	*85%*	*83%*	*88%*	*86%*
Combined	Fake Review	*86%*	*82%*	*93%*	*87%*

Finally, table10 provides outlines the performance indicators for three separate model's bigram, unigram, and combined unigram-bigram on a dataset specifically designed for identifying misinformation on disaster dataset. Within the framework of the Bigram model, using consecutive word pairs as features, a level of accuracy of 76% was achieved. This model demonstrated impressive accuracy of 73%, showing a significant percentage of correct positive predictions out of all positive predictions. Furthermore, it had a commendable recall rate of 91%, indicating its efficacy in accurately detecting a substantial proportion of true positive cases. The F1-Score, which represents the harmonic mean of accuracy and recall, was recorded as 81%.

Conversely, the Unigram model, which used individual words as features, had a superior overall accuracy of 80%. The performance of this model was well-balanced, with an accuracy of 77% and a recall of 91%. The F1-Score, which represents a balanced compromise between accuracy and recall, achieved a value of 84%.

The Combined Unigram-Bigram model, which incorporates both individual words and word pairs, demonstrated superior performance compared to both the Bigram and Unigram models.

The model had an accuracy rate of 81%, indicating a successful balance between precision (78%) and recall (91%). The F1-Score of the Combined model reached 84%, highlighting its capacity to use the advantages of both individual words and word pairs in feature representation.

Finally, all three models exhibited robust performance in identifying disinformation. Among them, the combined unigram-bigram model stood out as the most efficient, achieving a harmonious blend of individual word characteristics and contextual insights derived from word pairs.

Table 10: Result of Performance of different Featured based Model in Misinformation Dataset.

Model	Dataset	Accuracy	Precision	Recall	F1-Score
Bigram	Misinformation	*76%*	*73%*	*91%*	*81%*
Unigram	Misinformation	*80%*	*77%*	*91%*	*84%*
Combined	Misinformation	*81%*	*78%*	*91%*	*84%*

6.2. Discussion

The purpose of this work was to investigate the efficacy of three separate feature sets, namely Bigram, Unigram, and Combined Unigram-Bigram, in the context of three independent datasets that included information pertaining to cyberbullying, the detection of fake reviews, and the identification of misinformation.

In the dataset pertaining to cyberbullying, the Unigram approach demonstrated exceptional effectiveness, obtaining an accuracy rate of 73%. When compared to the other feature sets, our technique displayed much greater levels of accuracy, recall, and F1-Score. It is important to note that the Bigram approach, albeit having a lower accuracy rate of 68%, performed very well in recall, highlighting the importance of its efficiency in detecting good situations. Accuracy and recall were both taken into consideration in the Combined method, which offered a compromise. When selecting the proper feature set, the classification goals, the characteristics of the dataset, the distribution of the classes, and the computing resources that are available should all be taken into consideration.

Moving on to the Fake Review dataset, the Unigram approach fared better than the Bigram methodology, obtaining an accuracy of 85% with a precision, recall, and F1-Score that were all in good balance. A fair trade-off between precision and recall was established by the Combined Unigram-Bigram method, which achieved an accuracy of 86% during the course of study. It is remarkable that the Unigram method outperformed the Bigram strategy in terms of critical

parameters, which demonstrates the efficiency of the Unigram method in detecting malicious reviews.

In the dataset including misinformation, the Combined Unigram-Bigram model was shown to be the most effective, with an accuracy of 81%. The Bigram and Unigram models were both outperformed by this model, which properly balanced accuracy and recall. A balanced performance was also shown by the Unigram model, which had an accuracy of 80%. On the other hand, the Bigram model had outstanding accuracy and recall. The potential of the Combined model to offer a comprehensive representation of characteristics was shown by the fact that it used both individual words and word pairs.

The particular goals of the classification assignment should be taken into consideration while selecting the feature set that is found to be the most appropriate across all datasets. Comparatively, the Bigram strategy performed very well in terms of memory, whereas the Unigram technique consistently displayed higher overall performance. Achieving a balance between accuracy and recall was successfully accomplished by the use of the Combined method.

The outcomes of the research highlight how important it is to customize feature selection to the specific features of the dataset as well as the classification standards that are being pursued. It is possible that subsequent study might investigate the influence that more characteristics or sophisticated methodologies have on the performance of the model. In addition, it is essential to take into account the trade-offs that exist between accuracy and recall when one is dealing with real-world applications.

In conclusion, the findings of this study provide significant insights into the performance of various feature sets in a variety of classification tasks. These findings also provide a basis for making educated decisions on the selection of feature representations for text-based datasets.

CHAPTER SEVEN: CRITICAL REFLECTION

7.1. Critical Reflection

When I started working on this project, it was the beginning of my adventure into the world of machine learning and Python programming. When I first started working on this project, my understanding in these fields was inadequate, and the idea of navigating through the complexities of natural language processing and text mining looked intimidating to me. On the other hand, as the project progressed, I realized that I was progressively getting a fundamental grasp of the ideas of machine learning and growing skill in Python programming. My introduction into the field of data science was facilitated by the project, which offered me the opportunity to get practical experience that helped me understand difficult ideas and develop a set of skills that are applicable in real-world situations.

Furthermore, not only did this study provide me with the technical know-how I needed, but it also provided me with vital insights into the present situation of machine learning and artificial intelligence in the actual world. Through the process of investigating the identification of fraudulent activity on social media platforms, I was able to acquire a more nuanced comprehension of the ways in which these technologies are actively utilized to address modern concerns. The investigation of practical applications brought to light the relevance of ethical issues, the protection of personal data, and the ongoing development of models to keep up with the ever-changing dynamic settings of the internet.

In summary, this project has not only increased my technical skills in Python and machine learning, but it has also widened my understanding of the vital role that these technologies play in solving difficulties that are encountered in the real world. An excellent basis for future initiatives in the dynamic and ever-evolving area of data science and artificial intelligence, it serves as a testimony to the transformational potential of hands-on learning and provides a strong foundation for future endeavors in this subject.

As I investigated the use of natural language processing (NLP) and text mining for social media fraud detection, I discovered several advantages associated with models such as Naive Bayes, N-gram, and TF-IDF. The efficacy of these models was admirable, and they offered a systematic framework for detecting fraudulent activities in various domains. An example of this is the Naive Bayes classifier, which has proven to be a simple yet impactful instrument, providing a methodical and expeditious means of examining extensive datasets and thus making a substantial

contribution to the field of real-time fraud detection. The models' simplicity and efficiency render them exceptionally well-suited for managing the extensive and ever-changing terrain of social media data.

Nevertheless, a thorough analysis has identified certain aspects that necessitate enhancement. The complex nature of social media language may be oversimplified by the assumption of independence between features applied to Naive Bayes. In addition, it is worth noting that although N-gram models are effective at identifying local patterns, they may encounter difficulties when confronted with long-range dependencies, which could result in the omission of subtle nuances present in fraudulent expressions. These models may demonstrate limitations, for instance, when applied to the identification of sardonic remarks or culturally specific language that is indicative of fraudulent activities. This acknowledgment highlights the necessity for continuous progress to improve contextual comprehension and tackle these complexities in social media communication.

During this investigation, ethical questions become more prominent, highlighting the significance of responsible use of artificial intelligence. Important considerations include protecting the privacy of users, addressing the possibility of bias in training data, and considering the effects that fraud detection algorithms have on society. In the future, research activities should concentrate a greater emphasis on developing these models while keeping a close eye on ethical issues. It is necessary to continually develop and modify AI applications to remain ahead of new risks and societal expectations. This is because the nature of language trends on social media is always changing, and it is also vital that AI apps be created responsibly.

CHAPTER EIGHT: CONCLUSION AND FUTURE DIRECTION

8.1. Conclusion

This research attempted to conduct a comprehensive investigation into the use of Natural Language Processing (NLP) and Text Mining methods to tackle the widespread problem of fraud detection on social media platforms, which are in a state of continuing change. In order to uncover the hidden patterns and anomalies that are associated with fraudulent behaviour across a variety of social media platforms, it was essential to combine language analysis and data mining techniques.

The cyberbullying, fake review, and misinformation related to disaster from tweets databases were the subjects of the research we conducted, which comprised three distinct databases that were primarily focused on fraud. The information included in each database was particular to a certain aspect of fraudulent activities that occurred on social media. As a result of doing thorough study, we discovered unique insights that are associated with various domains, which has provided us with a greater understanding of the complexity of fraud in a variety of contexts. When it came to identifying fraudulent patterns, the models showed remarkable performance, with promising levels of accuracy, precision, recall, and F1-Score metrics throughout the process.

The comparative study that was carried out across a few different domains generated valuable information on the flexibility and limits of the text mining and natural language processing models. The complexity of fraud requires an understanding of the specific complications that are present within a given industry, even though quite a few characteristics have constantly shown their efficacy. It is clear from the findings of our study that there is a pressing need to adapt fraud detection techniques to the features that are associated with each social media site.

The paper offers specific recommendations for the successful implementation of these models in actual situations. During the deployment of a system, it is crucial to thoroughly assess aspects such as scalability, computational resources, and ethical implications. In addition, the ethical aspect of implementing these models requires dealing with problems of privacy, reducing bias, and promoting fairness to ensure responsible and equal practices in fraud detection.

It is critical that we prioritize on continuous progress and development as we plan the future course of our research. Subsequent study should focus on strengthening model durability, researching complex techniques, and examining innovative data sources. The ongoing effort to

get a complete understanding of fraud dynamics in the ever-changing social media ecosystem is a constant process, and this research lays the groundwork for future improvement in the field.

In summary, the combination of NLP and Text Mining has enormous promise for detecting fraud on social media platforms. This research not only contributes to the academic debate, but it also provides practical insights for professionals and politicians tasked with minimizing fraud in the ever-changing world of social media.

8.2. Recommendations

A few suggestions are provided to enhance the efficacy of our social media fraud detection effort. In the beginning, we can develop a system capable of comprehending sarcasm, quips, and the various methods in which individuals communicate online. Consequently, misunderstandings and false alarms will be diminished. Moreover, it is critical that our system is routinely updated. Our system requires updates in the same way that applications on your phone do to detect new tactics that fraudsters may employ. Further enhancing the intelligence of our system could involve incorporating user-generated media, such as images, audio, and videos, in addition to text. For example, if an individual uploads an image that appears dubious, our system may also acquire knowledge from that. Users should also be able to provide feedback on content that has been flagged, so that the system can gain knowledge from actual user interactions. Finally, we must ensure that our system is straightforward to comprehend. Imagine the hypothetical scenario where the system could provide an explanation for its suspicion of fraudulent activity. It would be akin to having a supportive assistant who consistently updates you on relevant developments. By implementing these suggestions, we can enhance the functionality of our system, increase its reliability, and strengthen user protection against online fraud.

8.3. Limitation of the Research

When it comes to understanding those delicate and often challenging parts of language on social media, our technique, which makes use of Natural Language Processing and Text Mining, may have difficulty. This is something that we need to bear in mind. For instance, when individuals communicate in a highly casual manner or use sarcasm, our approach may not be able to fully capture such subtleties. Imagine for a moment that someone may make a joke about something that seems like fraud; the system could misunderstand what they are saying. Since model is not always capable of having a good understanding of the context, it may raise false alarms or fail to

identify the actual problems. Online behaviour and fraudulent strategies are always evolving, which presents even another obstacle. To be able to recognize new tricks, our model has to be kept up to date, which might be challenging to do continuously. As a result, there is space for improvement in terms of understanding the way individuals communicate online and adjusting to the always shifting landscape of electronic fraud.

8.4. Future Scope of the Research

Future research in the field of an NLP with text mining approach for detection of fraud may prioritize improving the model's capacity to adapt to changing fraud trends. This can be achieved by using advanced deep learning techniques such as transformers, which are effective in capturing intricate temporal relationships. Another interesting path is cross-platform fraud detection, which aims to tackle linked fraudulent behaviors across several social media platforms. Studying user behavior analysis, namely by examining irregularities in interaction patterns, provides a tangible illustration to improve the accuracy of fraud detection. Furthermore, it is imperative to prioritize endeavors aimed at enhancing the comprehensibility of models and taking ethical factors into account, such as mitigating biases and safeguarding privacy. These measures are essential for the responsible implementation of models in practical situations. Implementing a standardized system for evaluating metrics and using benchmark datasets will enhance the ability to uniformly analyze different models for fraud detection.

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Appendix

Appendix I: Source Code for in Python Environment

```
#Importing all required libraries


import re

import nltk

from nltk.corpus import stopwords import string

import pandas as pds

import seaborn as sns

import matplotlib.pyplot as plot

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.feature_extraction.text import CountVectorizer

from nltk.tokenize import word_tokenize

from wordcloud import WordCloud, STOPWORDS

from sklearn.naive_bayes import MultinomialNB

from sklearn.model_selection import train_test_split

from nltk.stem import PorterStemmer, WordNetLemmatizer

from sklearn.metrics import accuracy_score, classification_report


#Download the necessary data

nltk.download('punkt')

nltk.download('wordnet')

nltk.download('stopwords')


# Reading the dataset CSV file

path = '/content/dataset.csv'

pds.read_csv(path)
```

1.Data Pre-Processing Steps

```
# Data Cleaning


def clean_tweet(tweet):
    if pds.isna(tweet):  # Checking for missing values
        return ""


    # Replace URLs with 'URL'
tweet = re.sub(r'http\S+|www\S+|https\S+', 'url', tweet, flags=re.MULTILINE)


    # Replacing mentions with 'User'
    tweet = re.sub(r'@\w+', 'user', tweet)


    # Removing punctuations and special characters
    tweet = re.sub(r'[^\w\s]', '', tweet)


    return tweet


# Apply the cleaning function to each dataframe
df['tweet_text'] = df['tweet_text'].apply(clean_tweet)


# Convert to lowercase


df['tweet_text'] = df['tweet_text'].str.lower() \
    .replace(to_replace=[r'http\S+|www\S+|https\S+', r'@\w+', r'[^\w\s]'],
             value=['url', 'user', ''], regex=True)
```

Tokenization

```
# Tokenizing the text through function.
def tokenizing_text(text):
    return word_tokenize(text)
```

```
# Applying the function to df
df['tweet_text'] = df['tweet_text'].apply(tokenizing_text)

# Removing stop words
Def stop_words_rmov(words, custom_stopwords=None):
   stop_words = set(stopwords.words('english'))
   if custom_stopwords:
      stop_words.update(custom_stopwords)
   words_filtered = [word for word in words if word.lower() not in stop_words]
   return words_filtered

# Define custom stopwords
custom_stopwords = ['url', 'mkr', 'user','rt', 'mkr', 'didn', 'bc', 'n', 'm', 'im', ']

# Applying stop words removal to each dataset
df ['tweet_text'] = df['tweet_text'].apply(
   lambda words: stop_words_rmov(words, custom_stopwords)
)

Stemming and Lemmatizing
# Performing stemming
def pfm_stemming(words):
   stemmer = PorterStemmer()
   word_stemmed = [stemmer.stem(word) for word in words]
   return word_stemmed

# Performing lemmatization
def pfm_lemmatization(words):
   lemmatizer = WordNetLemmatizer()
   words_lemmatized = [lemmatizer.lemmatize(word) for word in words]
   return words_lemmatized
```

```
# Using the function for stemming and lemmatization
df['tweet_text'] = df['tweet_text'].apply(pfm_stemming)
df['tweet_text'] = df ['tweet_text'].apply(pfm_lemmatization)

# Convert tokenized words to sentences for each dataset
df['tweet_text'] = df ['tweet_text'].apply(lambda x: [' '.join(x)])
df ['tweet_text'] = df ['tweet_text'][1:-1]
```

2.Data Analysis and Visualization

```
#Word Cloud for the tweets.
subset1 = df[df['label'] == 'type']
text = subset1.tweet_text.values

# Convert the list of sentences into a single string
text_string = " ".join(map(str, text))

# Define your list of custom stopwords
custom_stopwords = ['url', 'mkr', 'user']

# Add custom stopwords to the default set of stopwords
stopwords = set(STOPWORDS).union(custom_stopwords)

# Generate the WordCloud with custom stopwords
cloud1 = WordCloud(background_color='white', colormap="Dark2", collocations=False,
                   width=2000, height=1000, stopwords=stopwords).generate(text_gender_string)

# Plot the WordCloud
plot.figure(figsize=(10, 5))
plot.imshow(cloud1, interpolation='bilinear')
plot.title("Gender",fontsize=30)
plot.axis('off')
plot.show()
```

```
# Code for analyzing Top 10 most used words in Unigram and Bigram.

type = 'Label'
subset_df = df[df['label'] == 'type']

# Extracting sentences for the specific type of cyberbullying
corpus = subset_df['tweet_text']

# Drop rows with missing values in 'sentences'
corpus = corpus.dropna()

# Convert the list of sentences into a single string
corpus_text = " ".join(map(str, corpus))

# Ignoring words like 'user' and 'url'
stop_words_unigram = ['user', 'url', 'fuck', 'as','like','got','get']
stop_words_bigram = ['user', 'url', 'fuck', 'as','like','got','get']

# Unigram (Single Words)
unigram_vectorizer = CountVectorizer(stop_words=stop_words_unigram, max_features=10)
unigram_counts = unigram_vectorizer.fit_transform([corpus_text])
unigram_features = unigram_vectorizer.get_feature_names_out()
unigram_counts = unigram_counts.sum(axis=0).A1

# Ploting Unigram Results (Horizontal Bar Graph)
plot.figure(figsize=(12, 6))
plot.barh(unigram_features, unigram_counts, color='skyblue')
plot.title(f'Top 10 Most Used Unigram Words for {type}')
plot.xlabel('Count')
plot.ylabel('Unigram Words')
plot.show()
```

```
# Bigram (Two-word Phrases)
bigram_vectorizer = CountVectorizer(stop_words=stop_words_bigram, ngram_range = (2, 2),
max_features=10)
bigram_counts = bigram_vectorizer.fit_transform([corpus_text])
bigram_features = bigram_vectorizer.get_feature_names_out()
bigram_counts = bigram_counts.sum(axis=0).A1

# Plot Bigram Results (Horizontal Bar Graph)
plot.figure(figsize=(12, 6))
plot.barh(bigram_features, bigram_counts, color='salmon')
plot.title(f'Top 10 Most Used Bigram Phrases for {type}')
plot.xlabel('Count')
plot.ylabel('Bigram Phrases')
plot.show()
```

3.Feature Extraction

```
#Top 20 feature extraction for Uni-Gram and Bi-Gram.
# Droping rows with missing values in the 'tweet_text' column
df = df.dropna(subset=['tweet_text'])

# Converting each element to a string and combine them into a single string
combined_text = ' '.join(map(str, df['tweet_text']))

# Unigram TFIDF
Vectoriz_unigram = TfidfVectorizer()
X_unigram = Vectoriz_unigram.fit_transform([combined_text])

# Displaying feature names for Unigram TF-IDF
feature_names_unigram = unigram_vectorizer.get_feature_names_out()

# Creating visualization DataFrame
```

```
df_unigram = pds.DataFrame(data=X_unigram.toarray(), columns=feature_names_unigram)

# Getting the top 20 features for Unigram TF-IDF
top_features_unigram = df_unigram.sum().nlargest(20)

# Printing the top 20 features for Unigram TF-IDF (non-graphical)
print("Top 20 Features for Unigram TF-IDF:")
print(top_features_unigram)

# Plotting bar graph for the top 20 Unigram TF-IDF features
plot.figure(figsize=(12, 8))
top_features_unigram.plot(kind='bar')
plot.title('Top 20 Features for Unigram TF-IDF')
plot.xlabel('Feature')
plot.ylabel('TF-IDF Value')
plot.show()

# Bigram TF-IDF
Vectorizer_bigram = TfidfVectorizer(ngram_range=(2, 2))
X_bigram = Vectorizer_bigram.fit_transform([combined_text])

# Displaying feature names for Bigram TF-IDF
Names_feature = bigram_vectorizer.get_feature_names_out()

# Create visualization DataFrame
df_bigram = pds.DataFrame(data=X_bigram.toarray(), columns= Names_ feature)

# Getting the top 20 features for Bigram TF-IDF
top_features_bigram = df_bigram.sum().nlargest(20)

# Print the top 20 features for Bigram TF-IDF (non-graphical)
print("\nTop 20 Features for Bigram TF-IDF:")
```

```
print(top_features_bigram)


# Plot bar graph for the top 20 Bigram TF-IDF features
plot.figure(figsize=(12, 8))
top_features_bigram.plot(kind='bar')
plot.title('Top 20 Features for Bigram TF-IDF')
plot.xlabel('Feature')
plot.ylabel('TF-IDF Value')
plot.show()
```

4. Model Fitting and Evaluation

Unigram Approch

```
# Data Splitting into testing & training sets
X_train, X_test, y_train, y_test = train_test_split(df['tweet_text'], df['label'], test_size=0.2,
random_state=42)


# Unigram TFIDF
Vectorizer_unigram = TfidfVectorizer()
X_train_unigram = Vectorizer_unigram.fit_transform(X_train)
X_test_unigram = Vectorizer_unigram.transform(X_test)


# Initializing and training Naive Bayes classifier
nb_classifier = MultinomialNB()
nb_classifier.fit(X_train_unigram, y_train)


# Making predictions on the test set
predict_y = nb_classifier.predict(X_test_unigram)


# Model evaluation
acuracy = accuracy_score(y_test, predict_y)
rport = classification_report(y_test, predict_y)
```

```
print("Accuracy:", acuracy)

print("Classification Report:\n", rport)


# Confusion matrix creating

matrix = confusion_matrix(y_test, predict_y)


# Plot the confusion matrix

plot.figure(figsize=(8, 6))

sns.heatmap(conf_matrix, annot=True, fmt="d", cmap="Greens", xticklabels=['age', 'ethnicity', 'gender',
'not_cyberbullying', 'other_cyberbullying', 'religion'], yticklabels=['age', 'ethnicity', 'gender',
'not_cyberbullying', 'other_cyberbullying', 'religion'])

plot.title('Confusion Matrix')

plot.xlabel('Predicted Labels')

plot.ylabel('True Labels')

plot.show()


#Bigram Approch

y = df['label']

X = df['tweet_text']


# Spliting the data

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)


# Creating a TF-IDF vectorizer for features as bigram

Vectorizer_bigram = TfidfVectorizer(ngram_range=(2, 2))


# Transforming the training and testing data

X_train_bigram = Vectorizer_bigram.fit_transform(X_train)

X_test_bigram = Vectorizer_bigram.transform(X_test)


# Initializing and fit a Multinomial Naive Bayes classifier
```

```
nb_classifier = MultinomialNB()

nb_classifier.fit(X_train_bigram, y_train)


# Making predictions on the test set

predict_y = nb_classifier.predict(X_test_bigram)


# Evaluating the model

accuracy = accuracy_score(y_test, predict_y)

classification_rep = classification_report(y_test, predict_y)


# Print the results

print(f"Accuracy: {accuracy}")

print("Classification Report:\n", classification_rep)


# Creating a confusion matrix

conf_matrix = confusion_matrix(y_test, predict_y)


# Plotting the confusion matrix

plot.figure(figsize=(8, 6))

sns.heatmap(conf_matrix, annot=True, fmt="d", cmap="Reds", xticklabels=['age', 'ethnicity', 'gender',
'not_cyberbullying', 'other_cyberbullying', 'religion'], yticklabels=['age', 'ethnicity', 'gender',
'not_cyberbullying', 'other_cyberbullying', 'religion'])

plot.title('Confusion Matrix')

plot.ylabel('True Labels')

plot.xlabel('Predicted Labels')

plot.show()



#Combined Unigram and Bigram Approch

y = df['label']


X = df['tweet_text']
```

```
# Data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Creating a TF-IDF vectorizer for unigram features
Vectorizer_unigram = TfidfVectorizer()
X_train_unigram = Vectorizer_unigram.fit_transform(X_train)
X_test_unigram = Vectorizer_unigram.transform(X_test)

# Create a TF-IDF vectorizer for bigram features
Vectorizer_Bigram = TfidfVectorizer(ngram_range=(2, 2))
X_train_bigram = Vectorizer_Bigram.fit_transform(X_train)
X_test_bigram = Vectorizer_Bigram.transform(X_test)

# Combine unigram and bigram features
X_traincombined = hstack([X_train_unigram, X_train_bigram])
X_testcombined = hstack([X_test_unigram, X_test_bigram])

# Initialize and fit a Multinomial Naive Bayes classifier
nb_classifier = MultinomialNB()
nb_classifier.fit(X_train_combined, y_train)

# Predictions applying on test
predict_y = nb_classifier.predict(X_test_combined)

# Model evaluation
acuracy = accuracy_score(y_test, predict_y)
classification_rp = classification_report(y_test, predict_y)

# Print the results
print(f"Accuracy: {acuracy}")
print("Classification Report:\n", classification_rp)
```

```
# Confusion matrix creation

confmatrix = confusionmatrix(y_test, predict_y)


# Plotting the confusion matrix

plot.figure(figsize=(8, 6))

sns.heatmap(conf_matrix, annot=True, fmt="d", cmap="Blues", xticklabels=['age', 'ethnicity', 'gender',
'not_cyberbullying', 'other_cyberbullying', 'religion'], yticklabels=['age', 'ethnicity', 'gender',
'not_cyberbullying', 'other_cyberbullying', 'religion'])

plot.title('Confusion Matrix')

plot.xlabel('Predicted Labels')

plot.ylabel('True Labels')

plot.show()
```